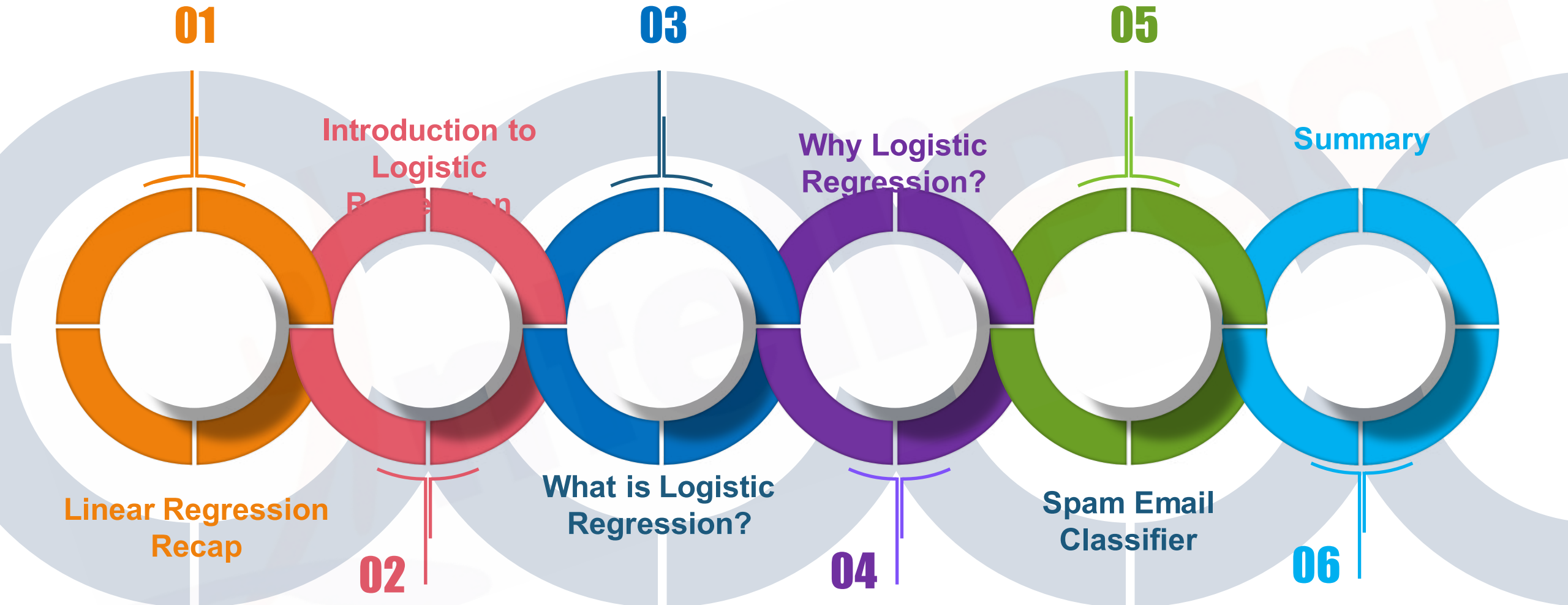


Logistic Regression Algorithm



Agenda for Today's Session



Look back at Linear Regression

Linear Regression (Recap)



Hey Josh! Help me find a property with bigger garden area for “X” bucks.

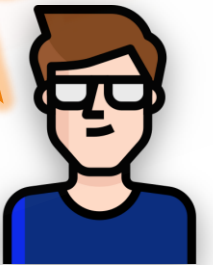


Hey Lauren! Sure, I will help you.

Linear Regression (Recap)



Hey. Can you do something about it?

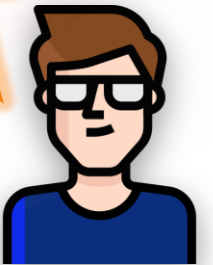


Sure...

Linear Regression (Recap)



Hey. Can you do something about it?

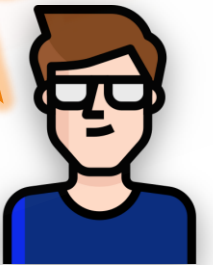


So,
Spending " X " bucks
will get her a
property of area " Y "

Linear Regression (Recap)



How did you find out?



Simple linear regression.

Linear Regression (Recap)

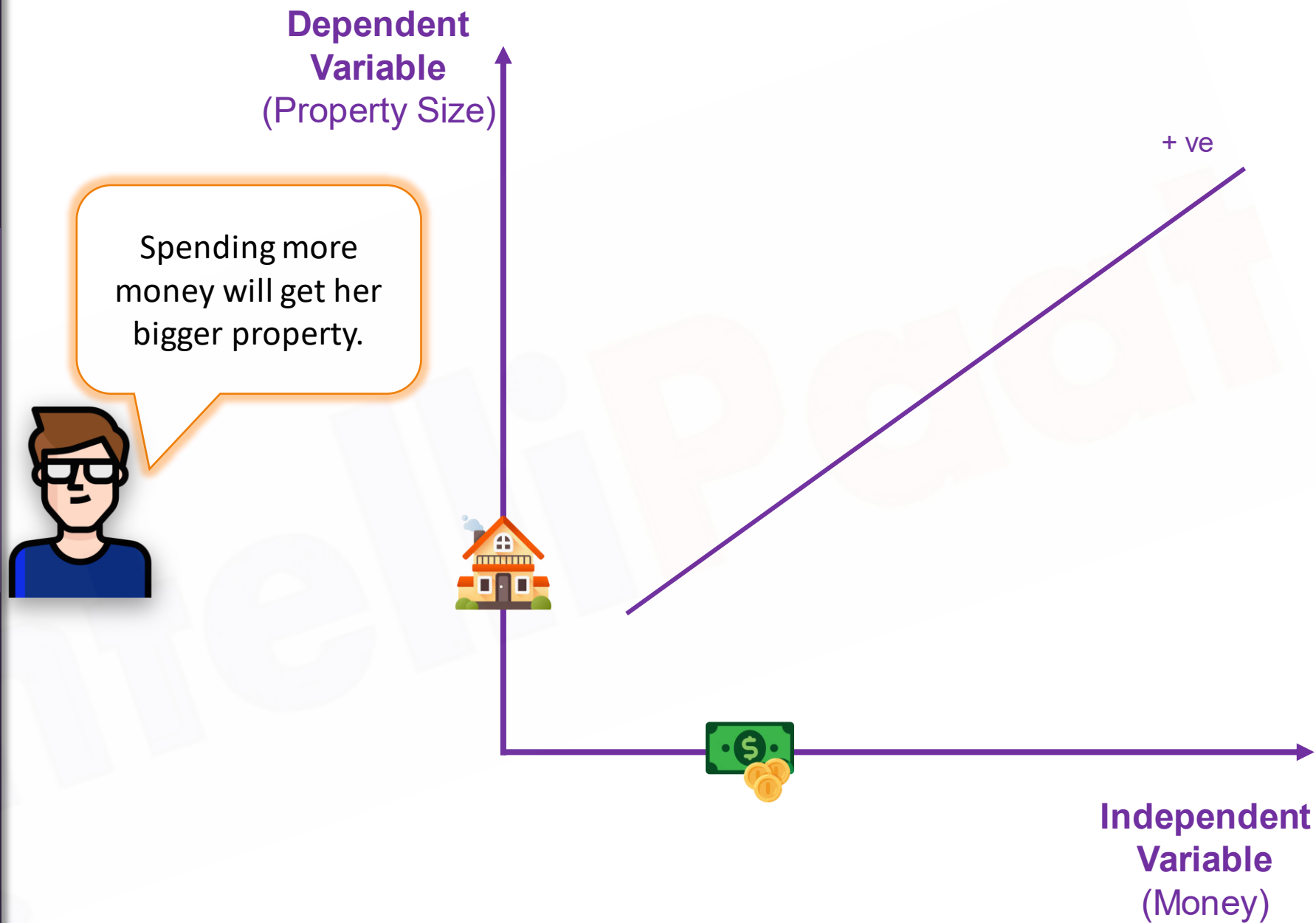
A cartoon illustration of a man with a brown beard and mustache, wearing a blue t-shirt.

How did you find out?

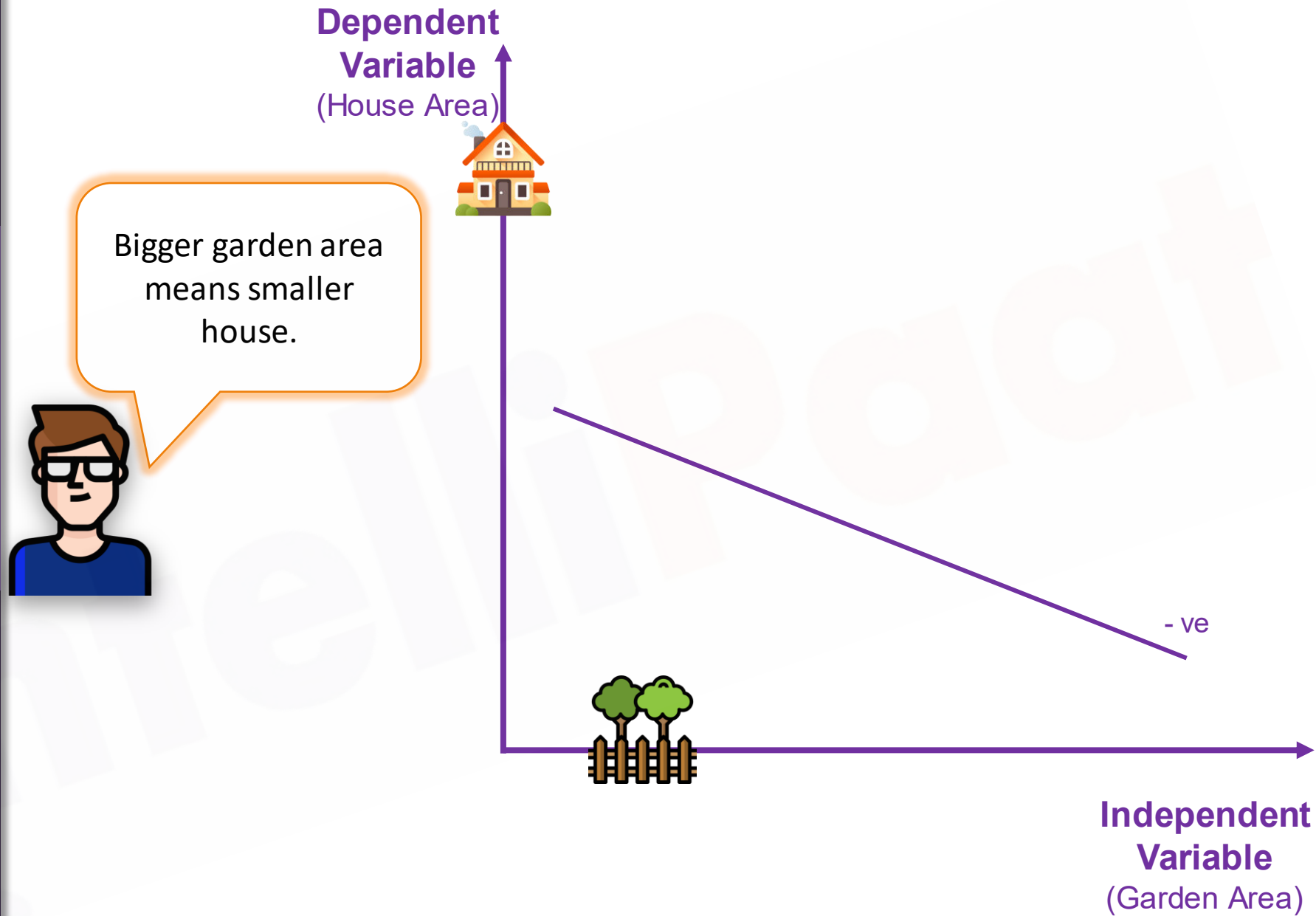
A cartoon illustration of a man with brown hair and glasses, wearing a blue t-shirt.

Let me show you how.

Linear Regression (Recap)

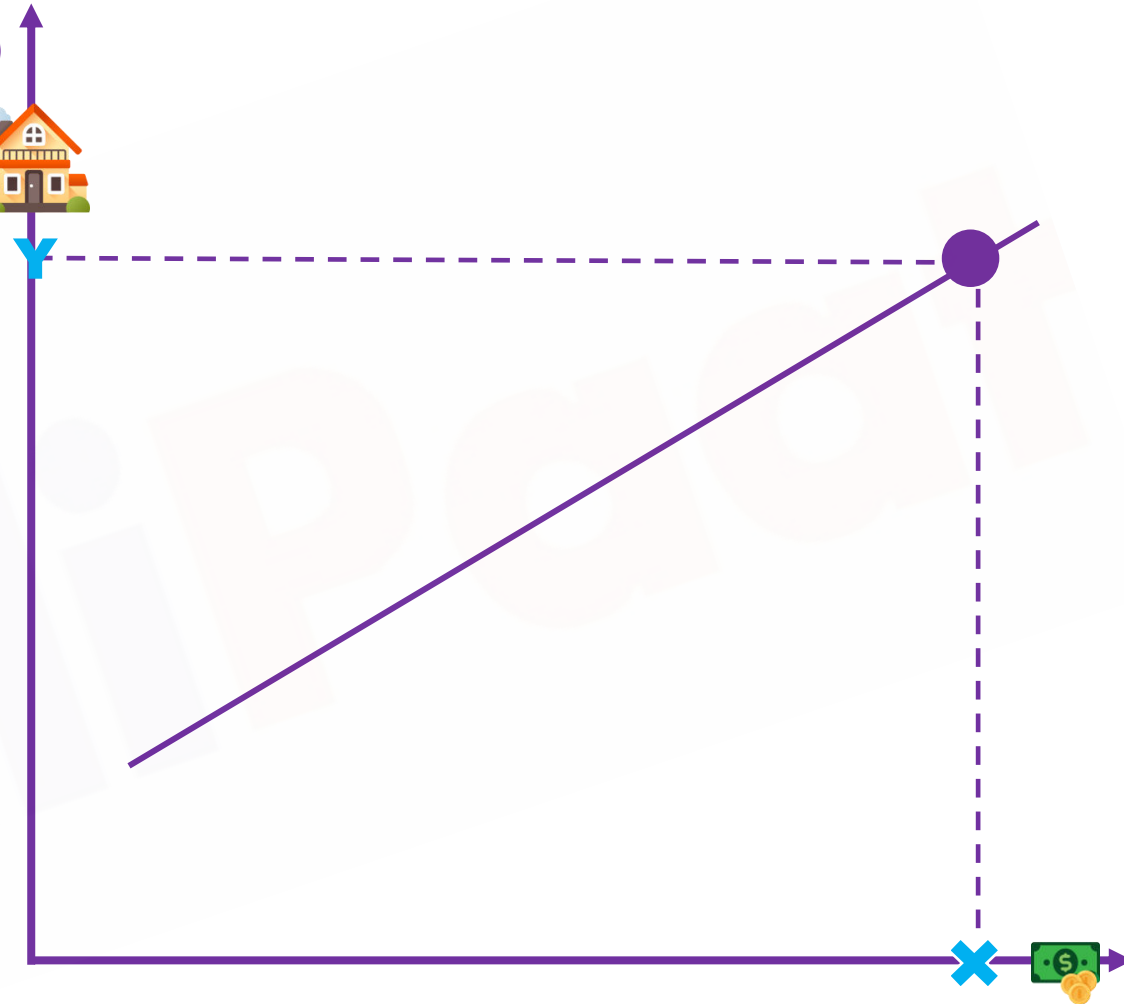


Linear Regression (Recap)





Dependent
Variable
(Property Area)



What he can say



If Lauren spends “X” amount of money, she can buy a property of area “Y”.

What he can't say



Will the property have a good neighbourhood or not?
Will the location be noiseless suburb or a bustling city?

Introduction to Logistic Regression



Will the property have a good neighbourhood?



Will it rain tomorrow or not?



Is the mail spam or not?

Classification Problems



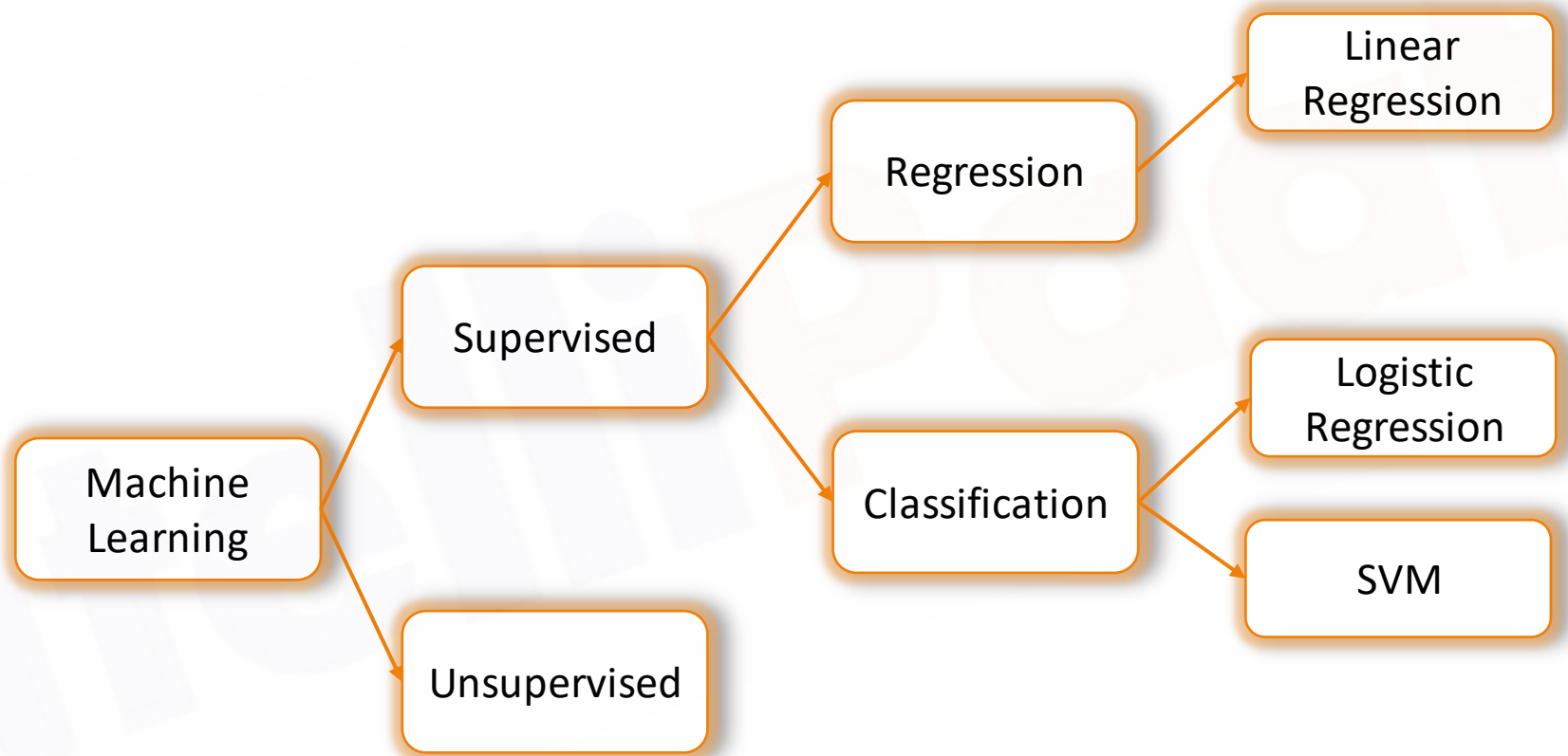
Linear Regression cannot answer



Logistic Regression comes into the picture

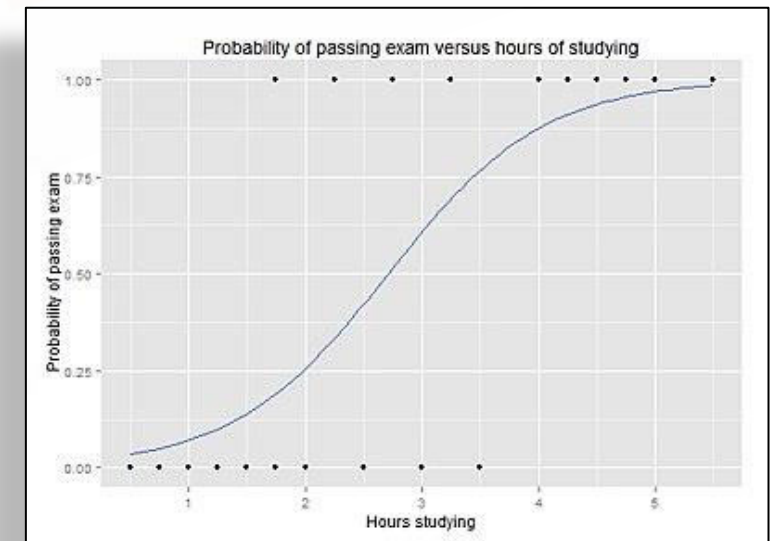


Introduction to Logistic Regression



What is Logistic Regression?

- A statistical classification model
- Deals with categorical dependent variables
- Could be binary or dichotomous
- Could be multinomial
- Takes both continuous and discrete input data



Why Logistic Regression ?

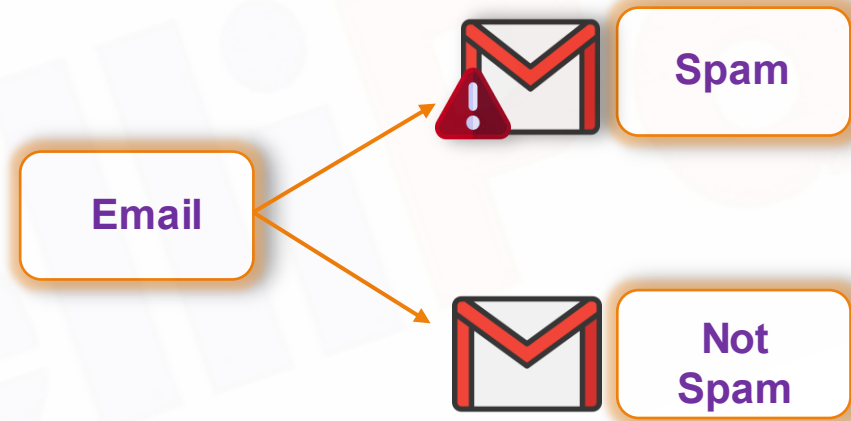
- Tool for applied statistics and discrete data analysis
- Gives outcome in terms of probability
- In-turn helps in classifying the given data



Let's understand this with an example

Spam Email Classifier

Let us explore with an example of Spam email classifier.



Spam Email Classifier

Steps:

- Understanding the variable
- Plot the labeled data
- Draw regression curve
- Find out the best fitted curve using Maximum Likelihood Estimator(MLE)

Spam Email Classifier

Let's get started.



STEP:1

Define the variable

Plot labeled data

Draw regression line

Find out the best using MLE

Independent variable:

- Count of spam words

Most common spam words:

- Buy
- Get paid
- Guarantee
- Winner
- Unlimited



Bag of spam words

STEP:1

Define the variable

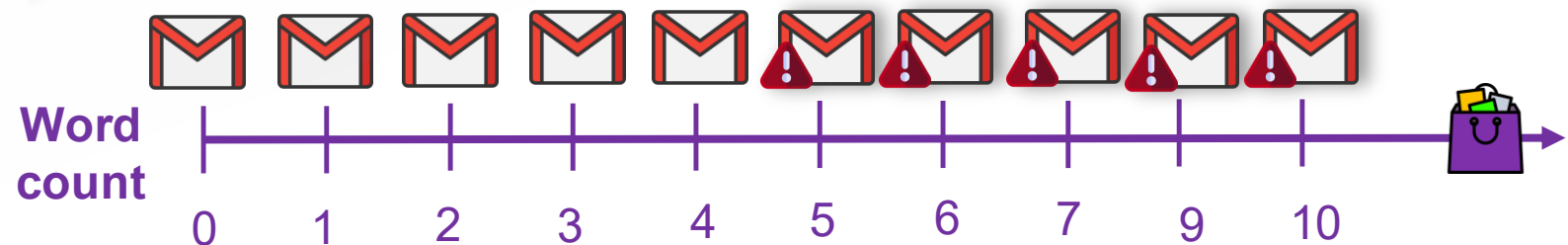
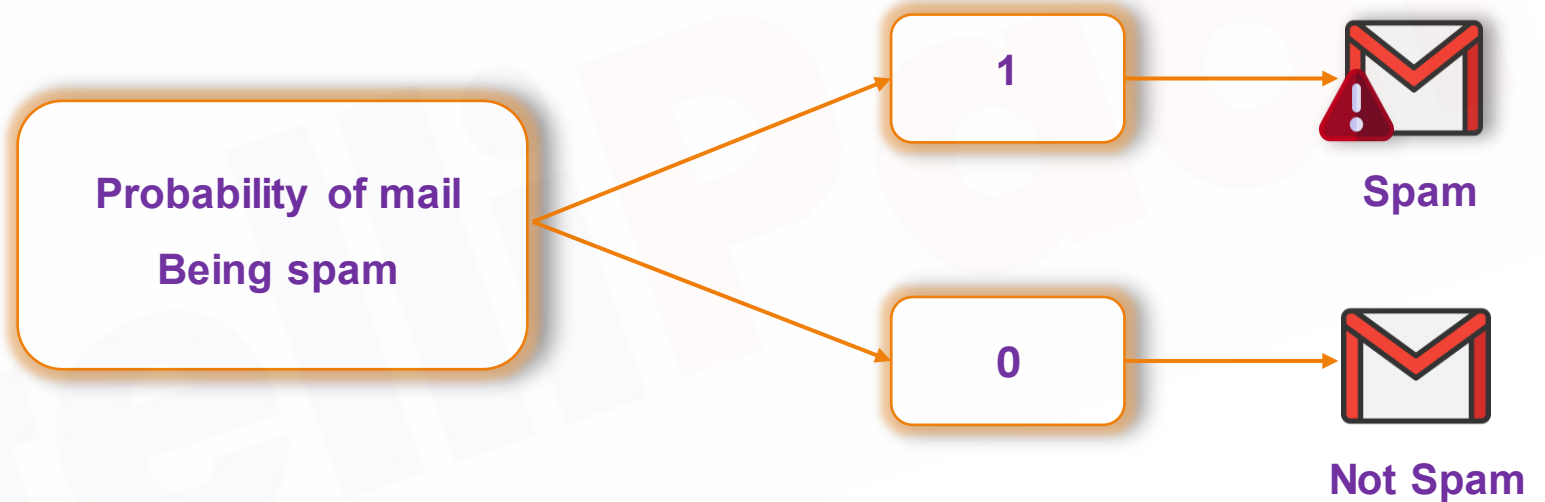
Plot labeled data

Draw regression line

Find out the best using MLE

Dependent variable:

- Probability of mail being spam.
- Binary or dichotomous



STEP:2

Define the variable

Plot labeled data

Draw regression line

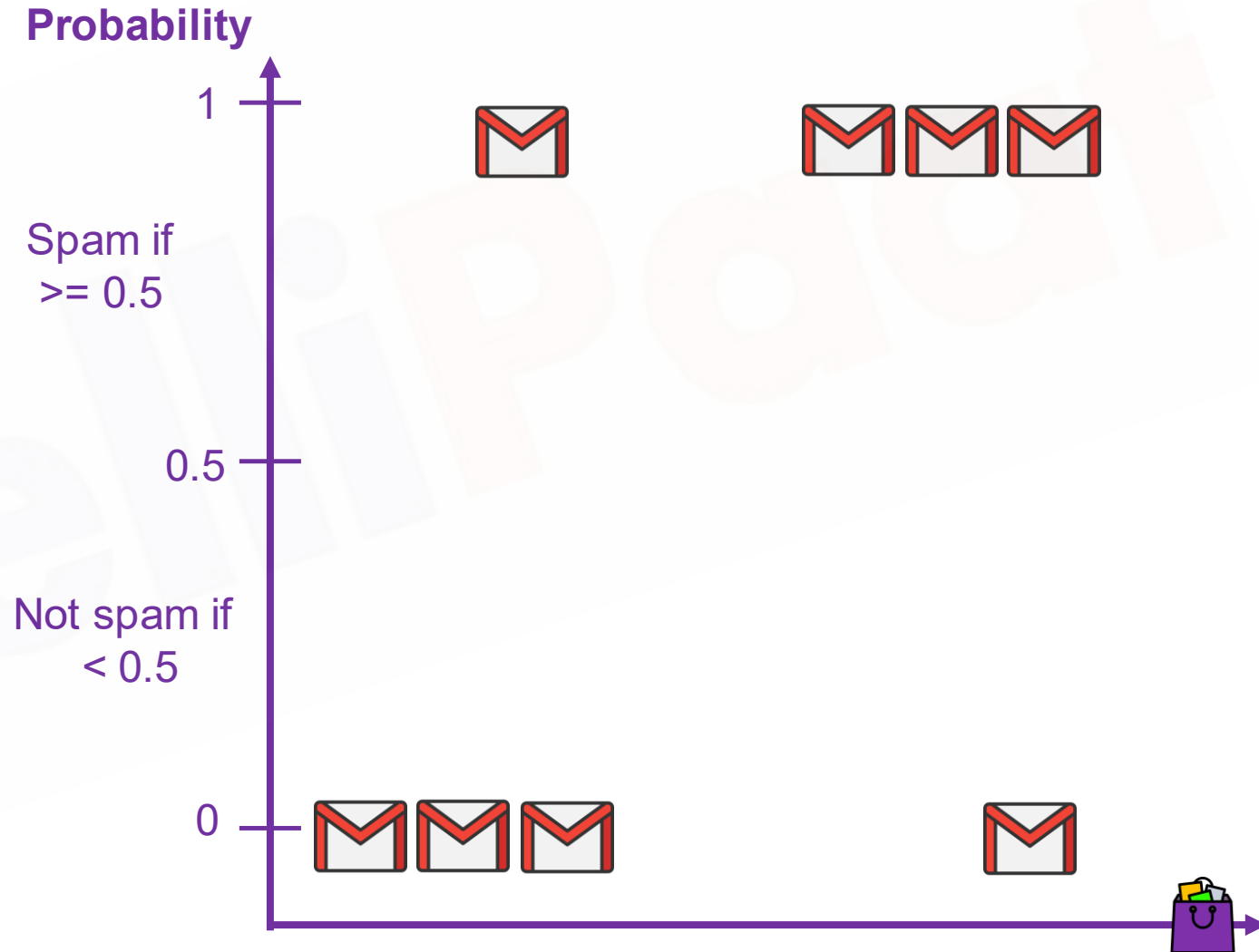
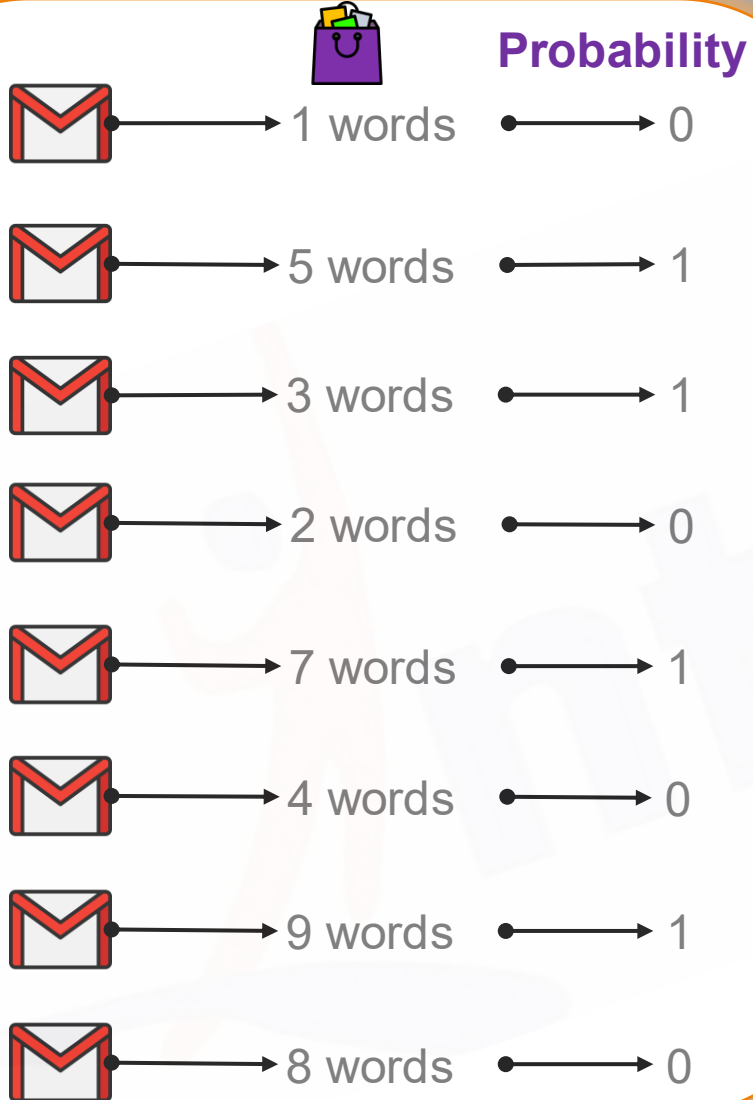
Find out the best using MLE

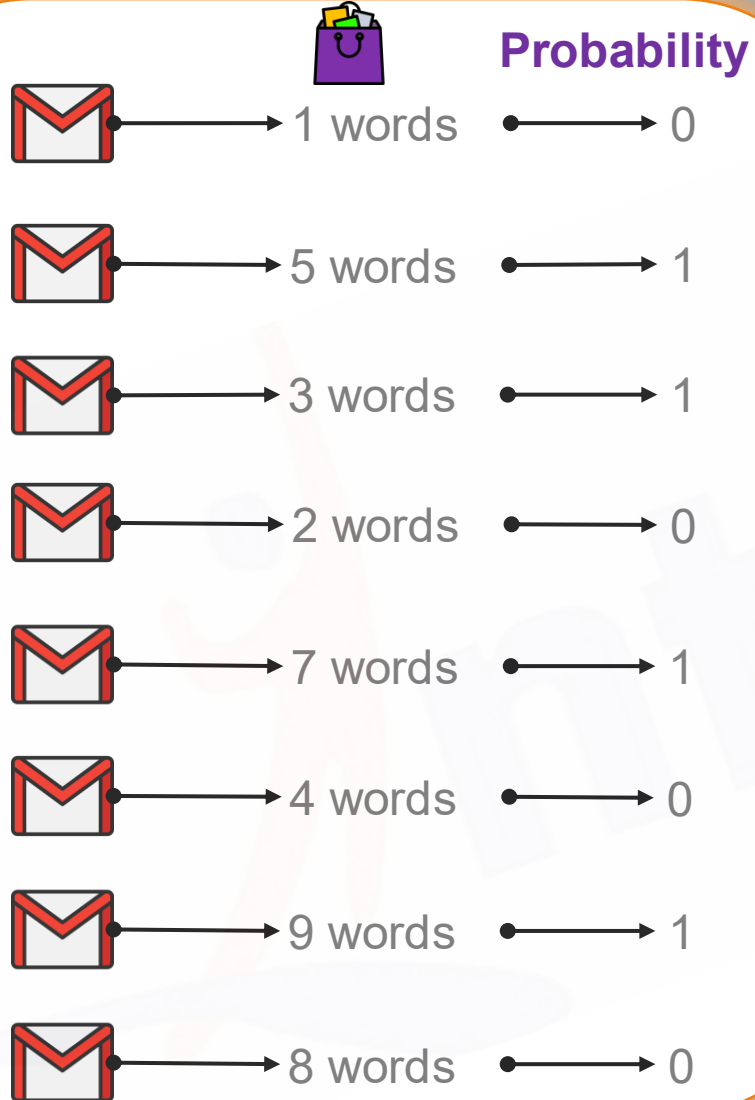
Probability		
	→ 2 words	→ 0
	→ 6 words	→ 1
	→ 8 words	→ 1
	→ 1 words	→ 0
	→ 7 words	→ 1
	→ 3 words	→ 0
	→ 9 words	→ 1
	→ 8 words	→ 0

Pre-labeled Dataset

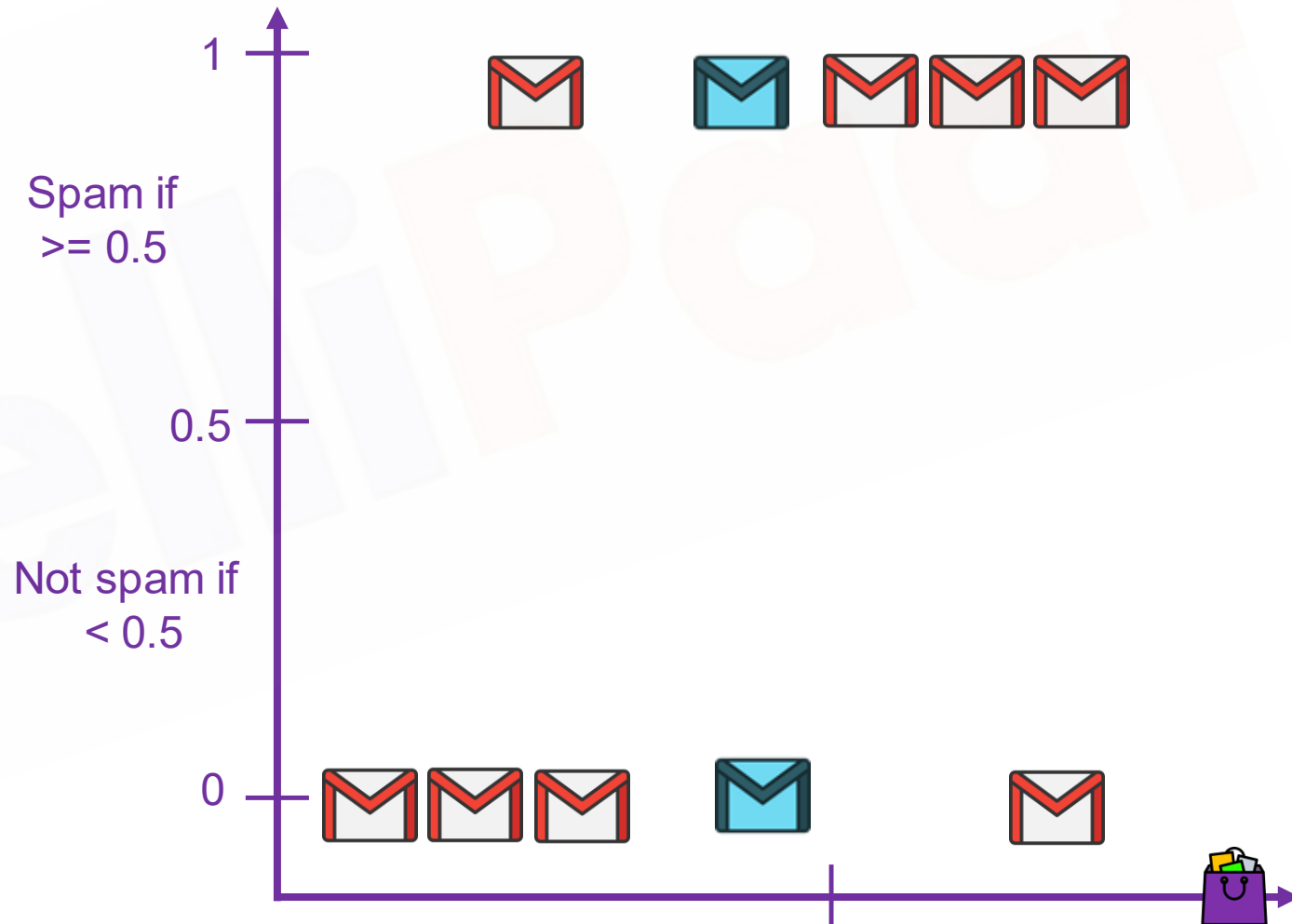
		Probability
	1 words	0
	5 words	1
	3 words	1
	2 words	0
	7 words	1
	4 words	0
	9 words	1
	8 words	0





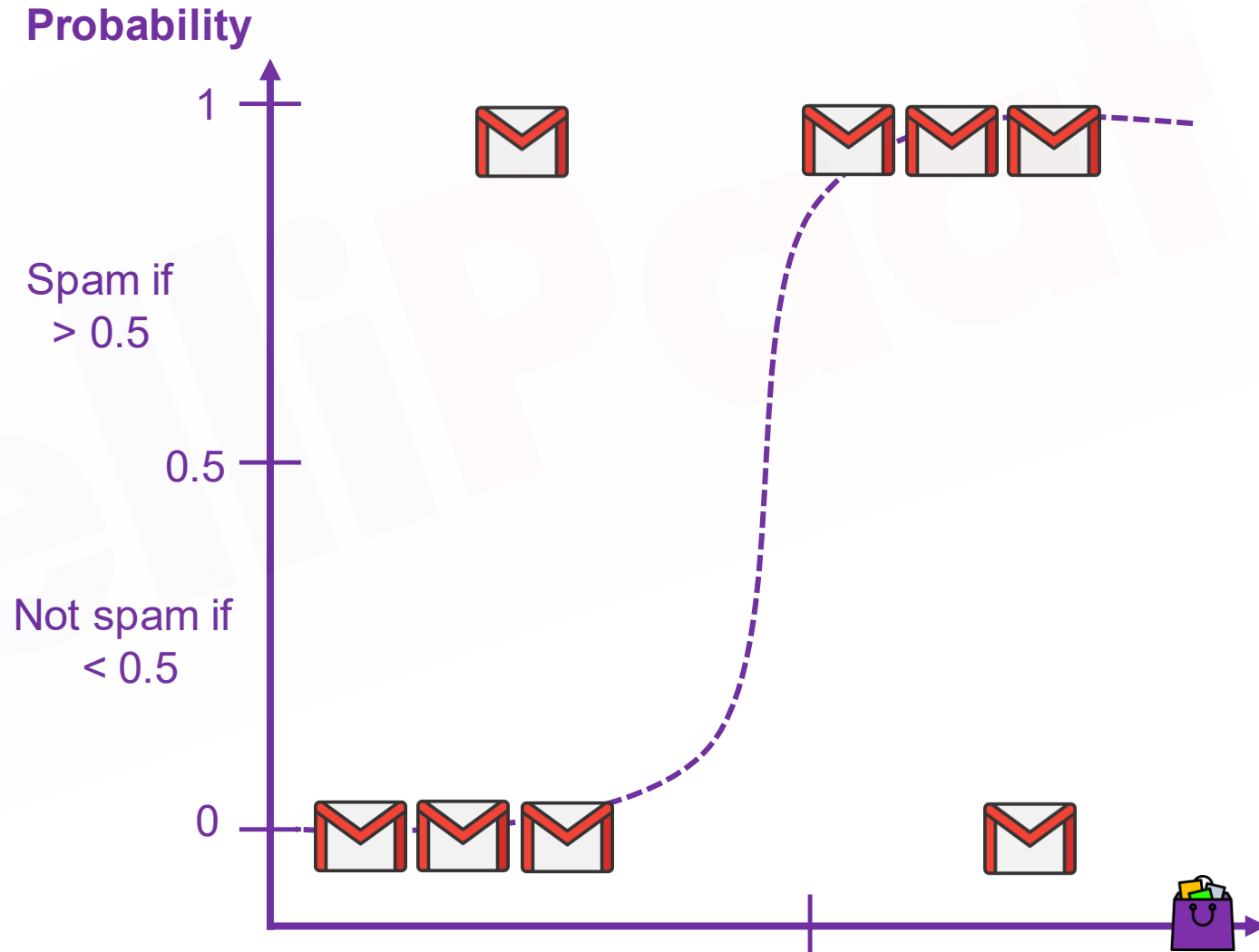


Probability

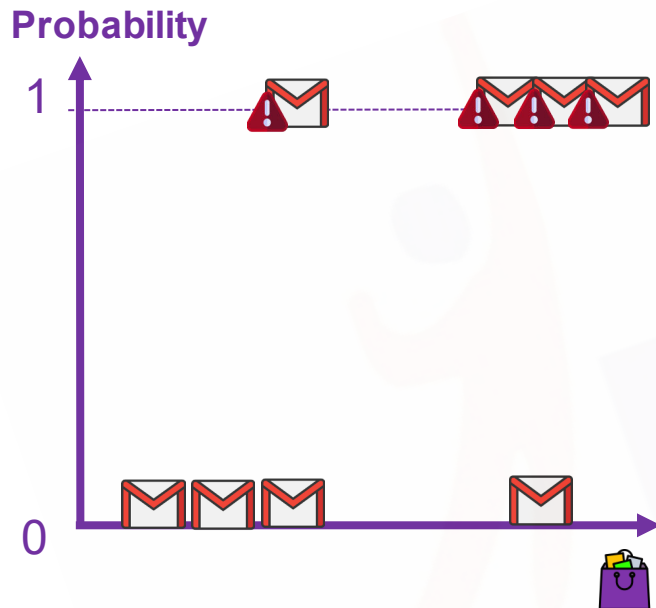


- Need to find a regression curve which would be the best fit
- That would be our logistic regression curve

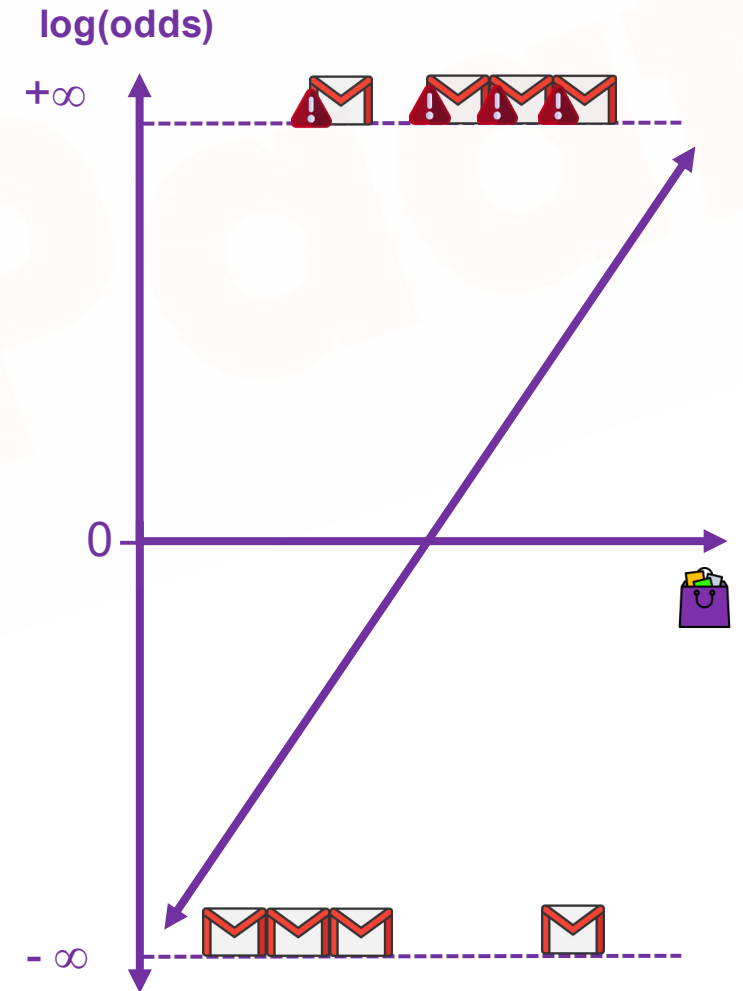
But how do we find the best fit?



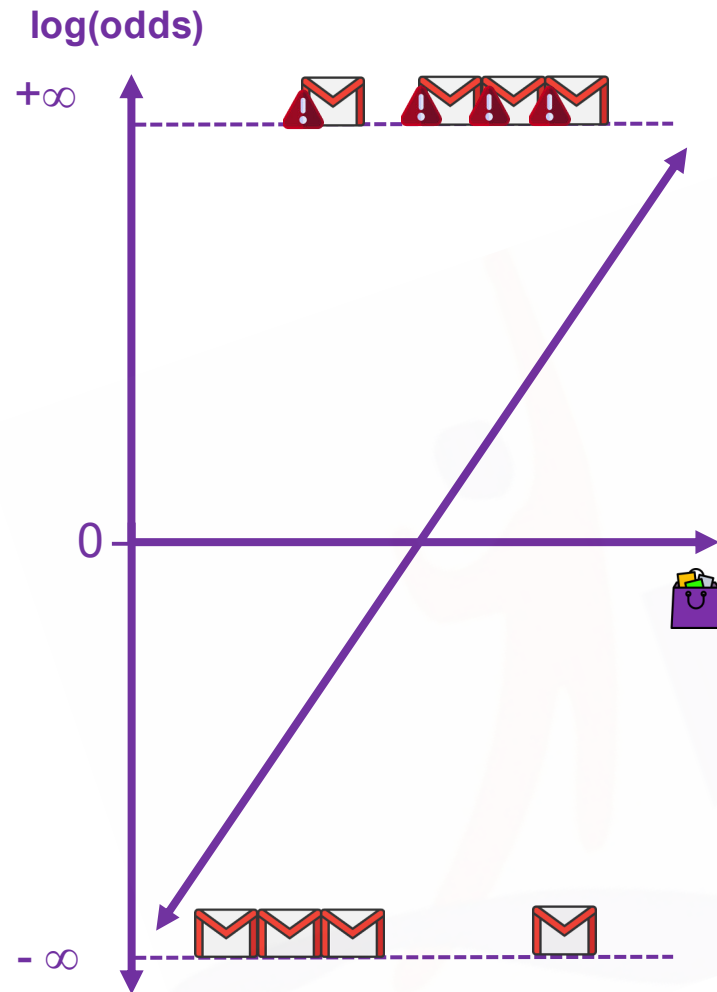
How?



log(odds)

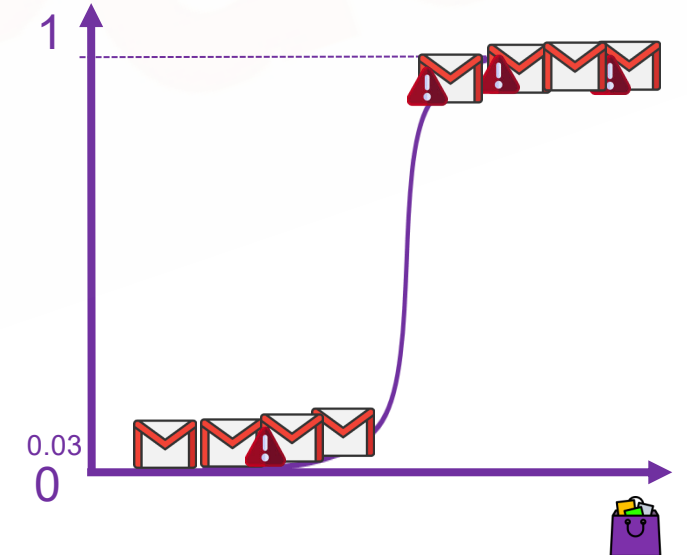


How?



Sigmoid
Function

Probability of
mail being spam



How?



STEP:3

Define the variable

Plot labeled data

Draw regression line

Find out the best using MLE

What does $\log(\text{odds})$ mean?



STEP:3

Define the variable

Plot labeled data

Draw regression line

Find out the best using MLE

Probability vs odds



- This guy went fishing 5 times a week
- Caught a fish 2 times
- Failed to catch 3 times.

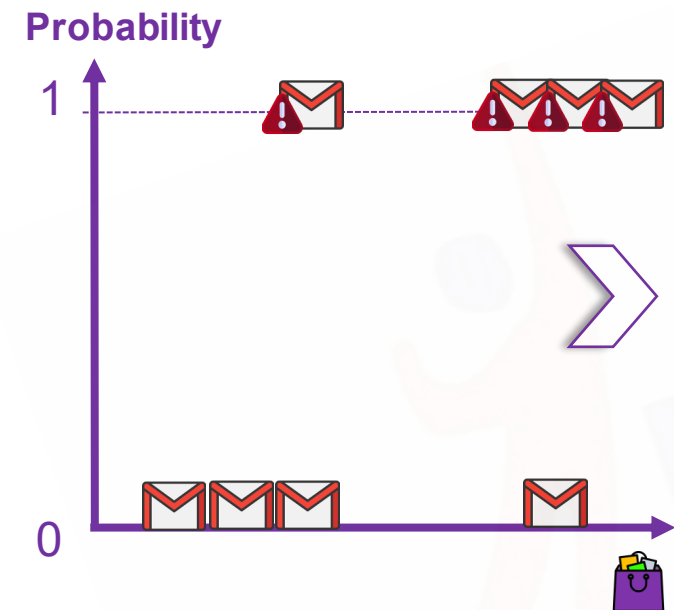
What is the probability and odds of getting a Fish for dinner?

$$\text{Probability} = \frac{\text{Chances for}}{\text{Total chances}} = \frac{2}{5}$$

$$\text{Odds} = \frac{\text{Chances for}}{\text{Chances Against}} = \frac{2}{3}$$







$$\log(\text{odds}) = \log\left(\frac{P(\text{Spam})}{1-P(\text{Spam})}\right)$$

$$\log(\text{odds}) = \log\left(\frac{1}{1-1}\right)$$

$$\log(\text{odds}) = \log\left(\frac{1}{0}\right) \rightarrow +\infty$$

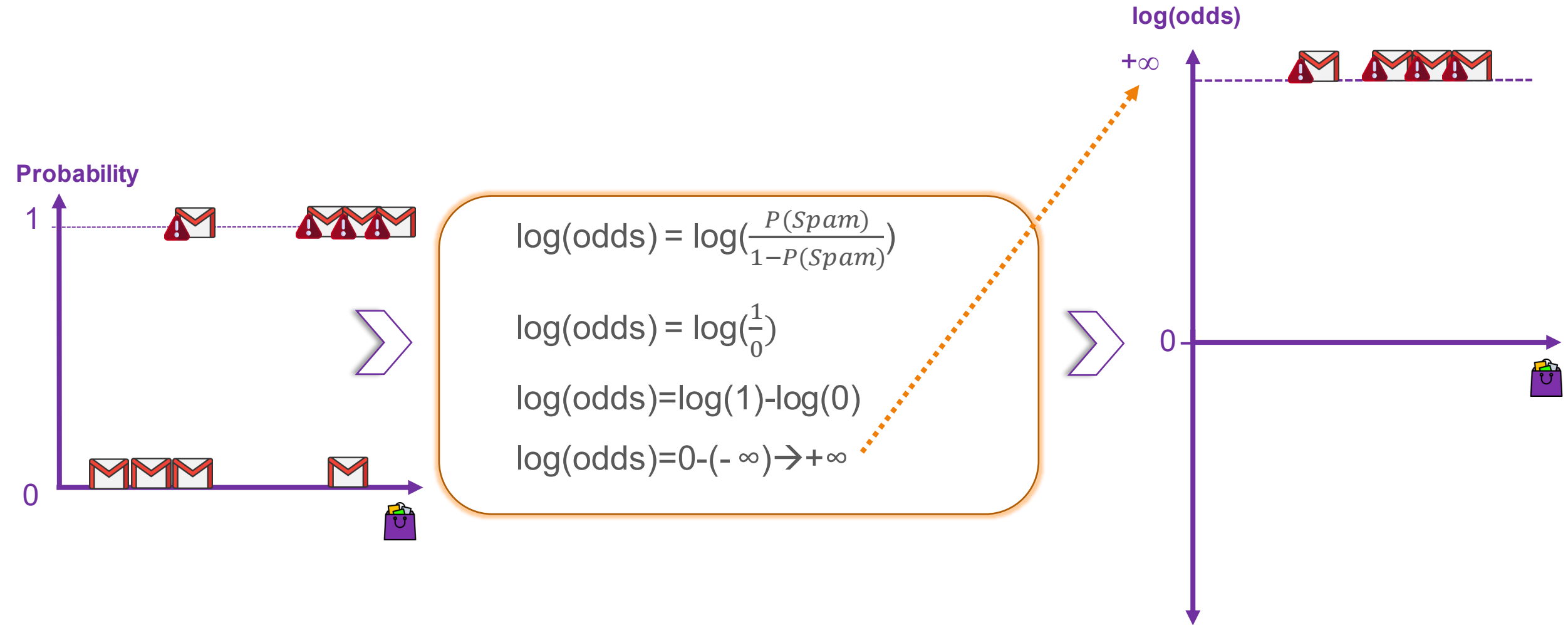
$$\log_b 0 = c$$

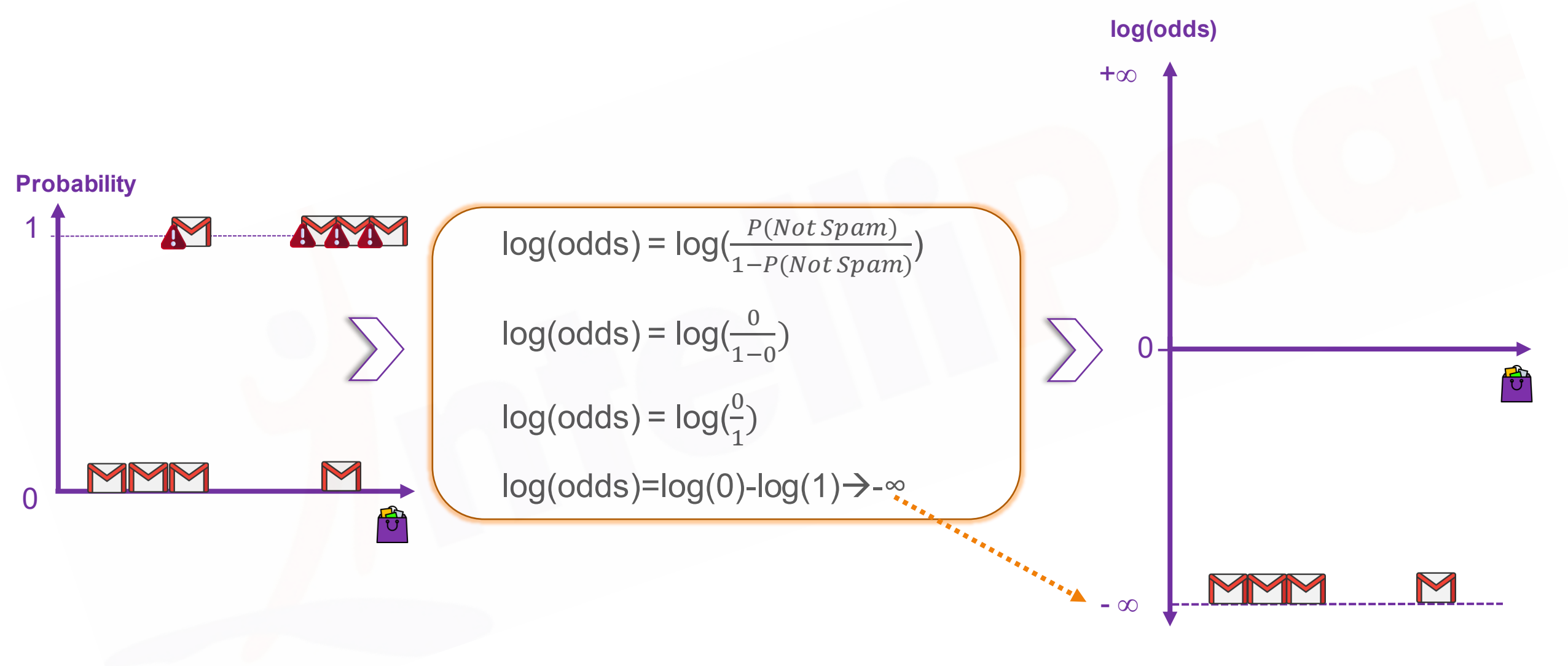
$$\Rightarrow 0 = b^c \text{-----}(1)$$

So, for equation(1) to be true.

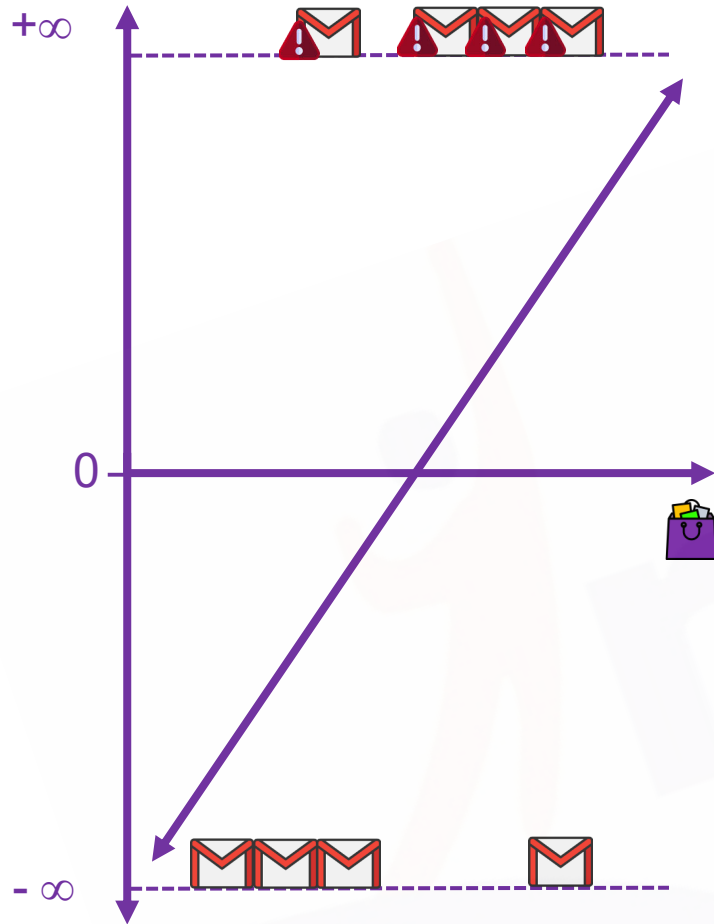
$$\text{If } b < 1 \Rightarrow c \rightarrow -\infty$$

$$\text{If } b > 1 \Rightarrow c \rightarrow +\infty$$

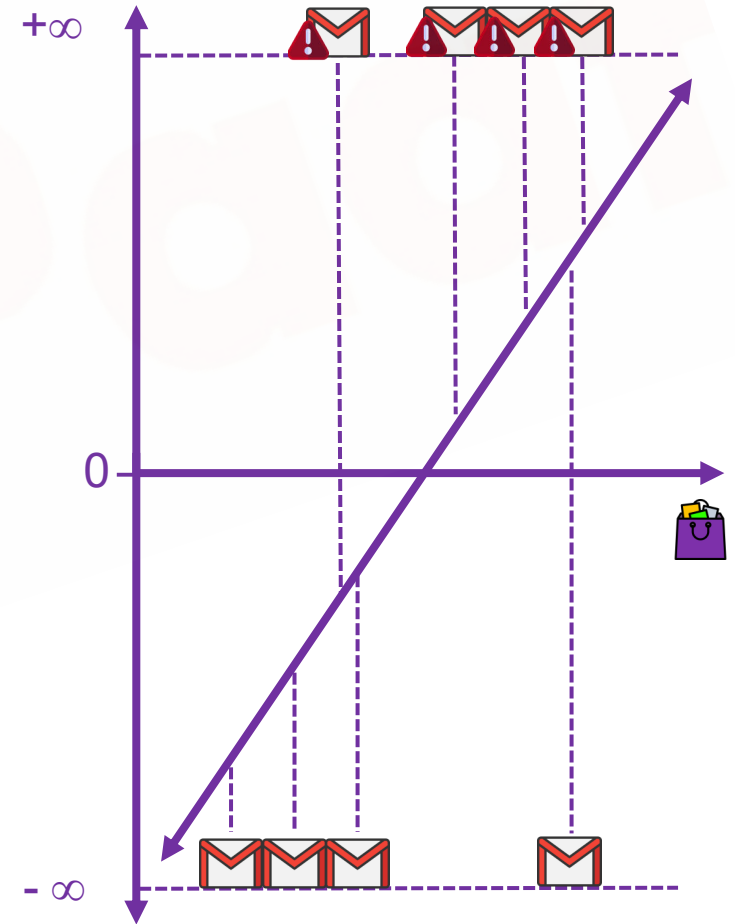




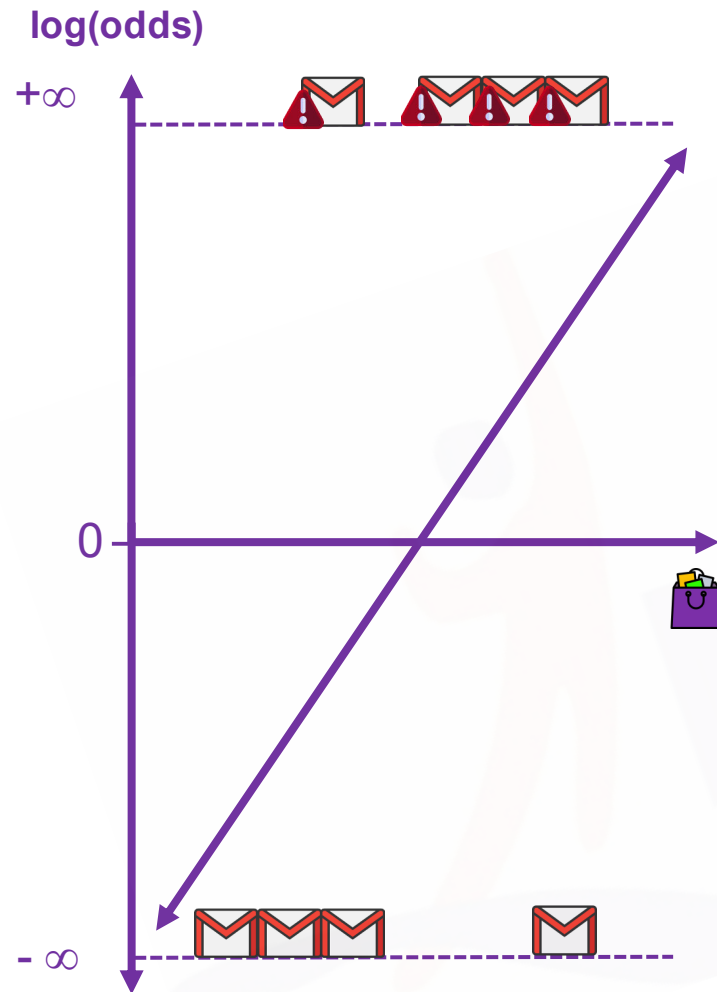
log(odds)



log(odds)

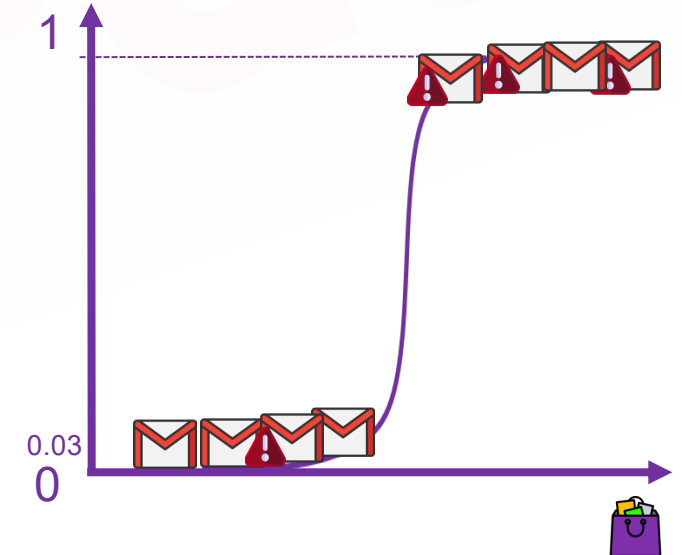


How?



Sigmoid
Function

Probability of
mail being spam



STEP:3

Define the variable

Plot labeled data

Draw regression line

Find out the best using MLE

What does Sigmoid Function mean?



STEP:3

Define the variable

Plot labeled data

Draw regression line

Find out the best using MLE

Sigmoid Function:

Sigmoid function is the standard logistic function

$$\text{Logistic function} = \frac{L(e^{k(x-x_0)})}{1+e^{k(x-x_0)}}$$

Here,

L – Curve's maximum value

k – Steepness of the curve

x0 – x value of Sigmoid midpoint

$$\text{Sigmoid function} = \frac{e^x}{1+e^x}$$

Here,

k=1

x0=0

L=1

STEP:3

Define the variable

Plot labeled data

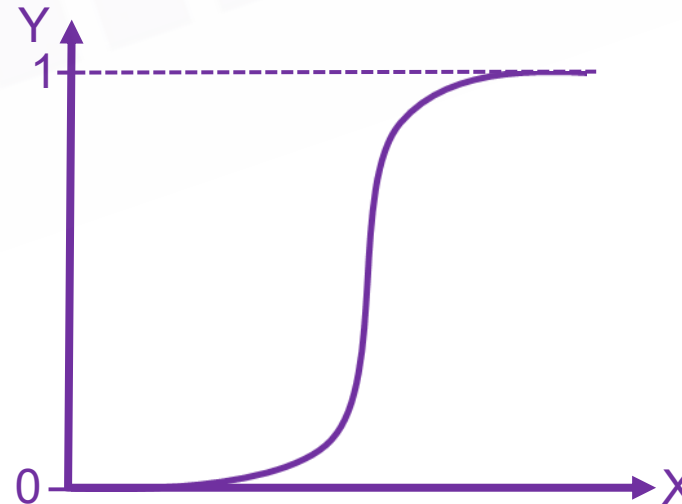
Draw regression line

Find out the best using MLE

Sigmoid Function:

$$\text{Sigmoid function} = \frac{e^x}{1+e^x}$$

- S' shaped curve.
- Sigmoid curve has a finite limit of:
 - '0' as x approaches $-\infty$
 - '1' as x approaches $+\infty$



STEP:3

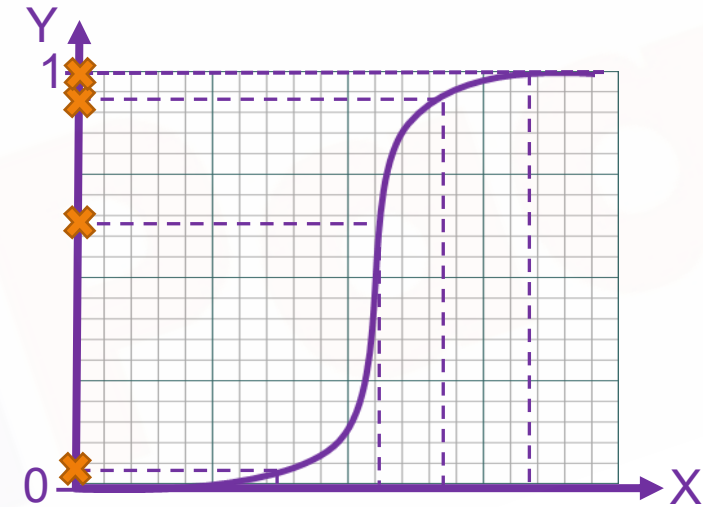
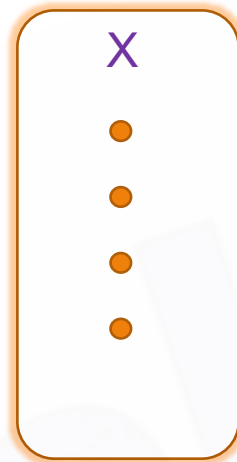
Define the variable

Plot labeled data

Draw regression line

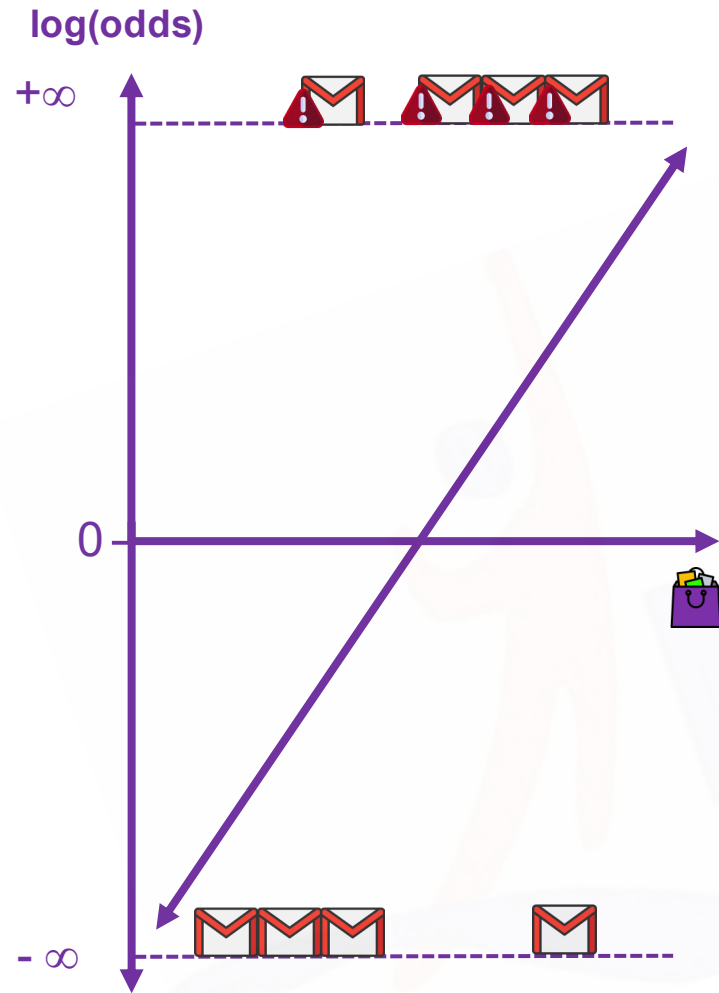
Find out the best using MLE

Sigmoid Function:

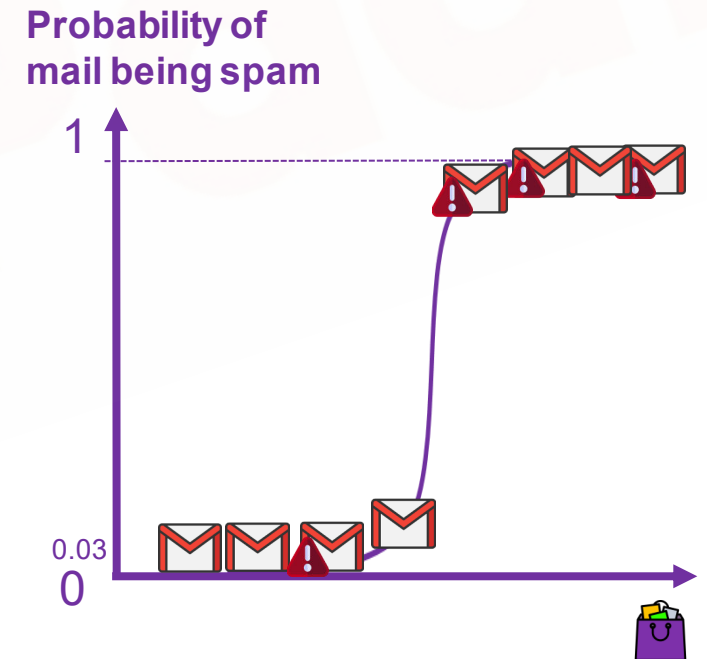


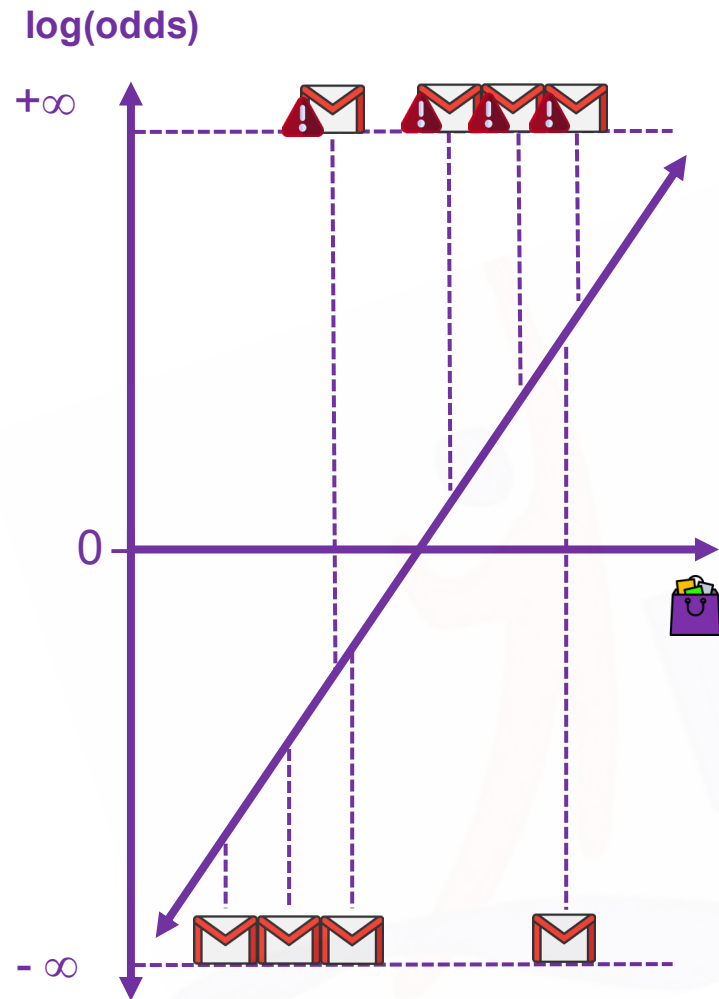
- Takes any real-valued number and maps into a value between 0 and 1.
- Helpful while solving classification problems

How?



Sigmoid Function

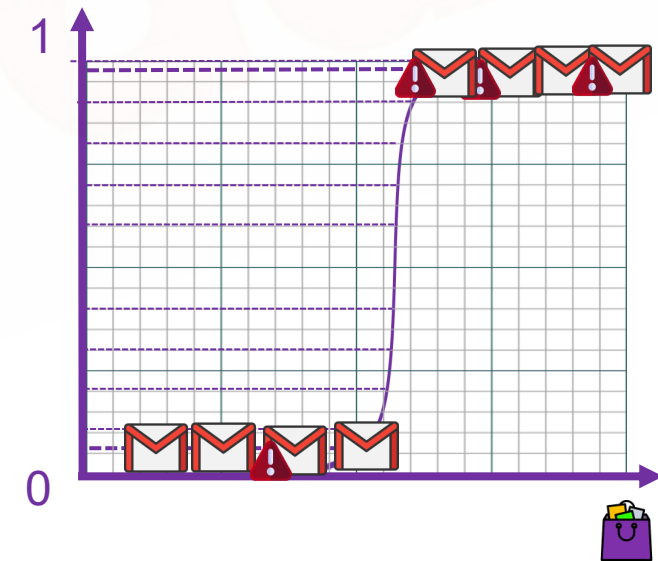


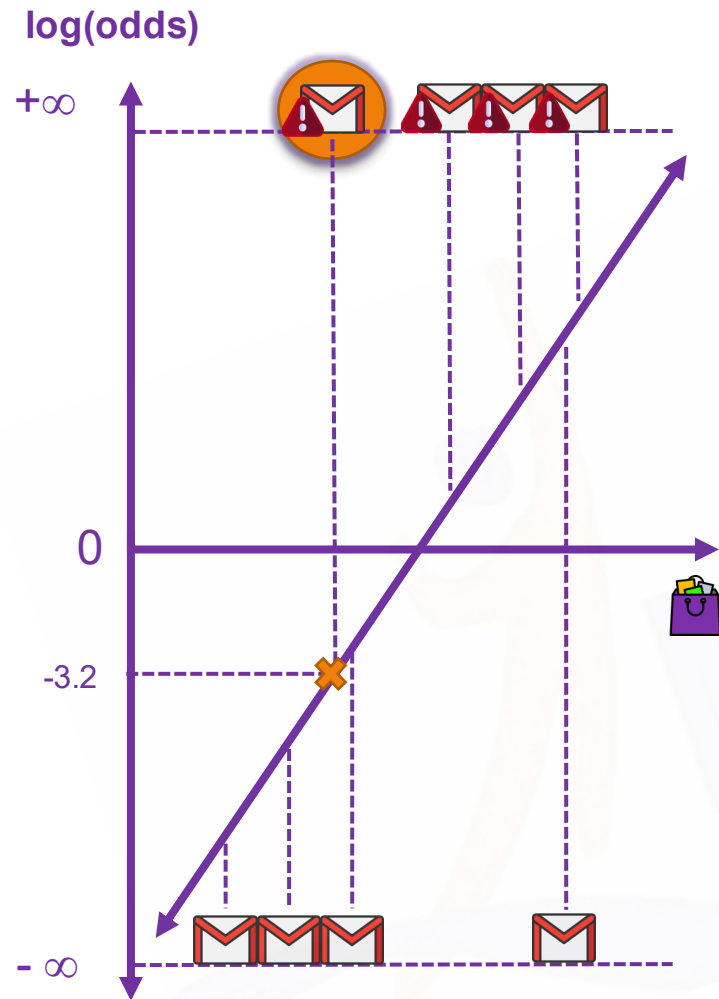


$$P = \frac{e^{\log(\text{odds})}}{1 + e^{\log(\text{odds})}}$$



Probability of mail being spam

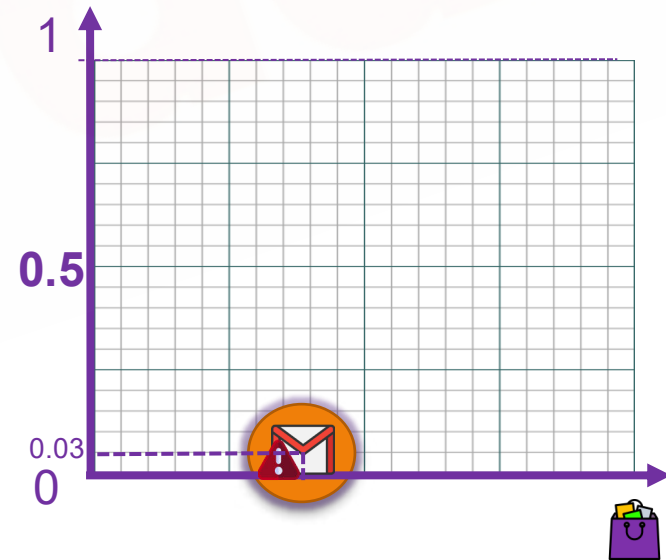


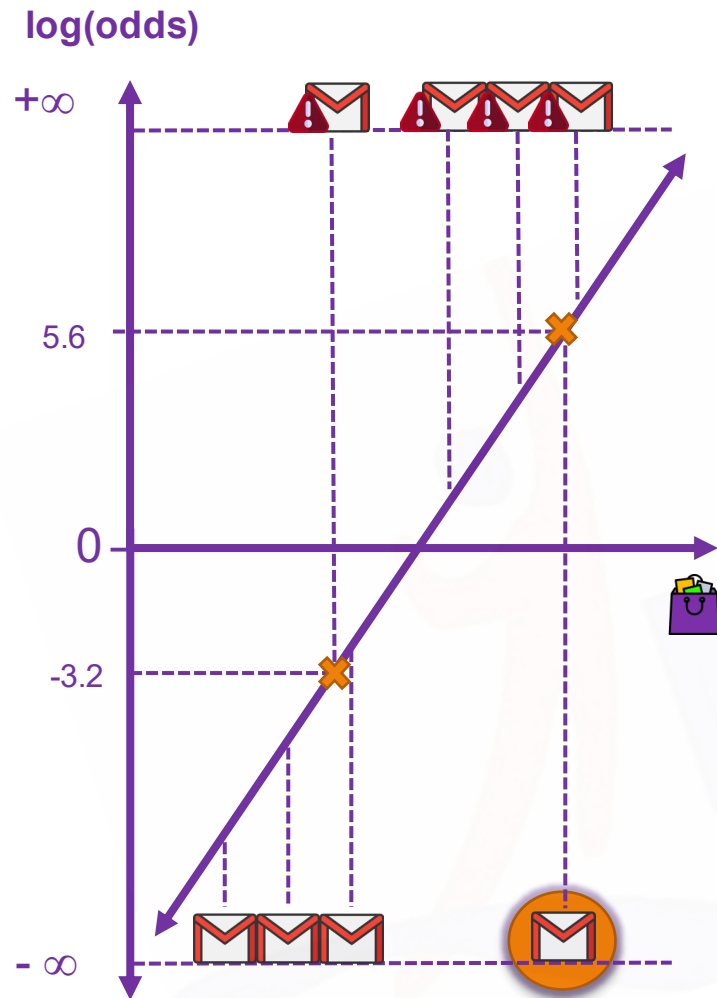


$$P = \frac{e^{\log(-3.2)}}{1 + e^{\log(-3.2)}} = 0.03$$



Probability of mail being spam

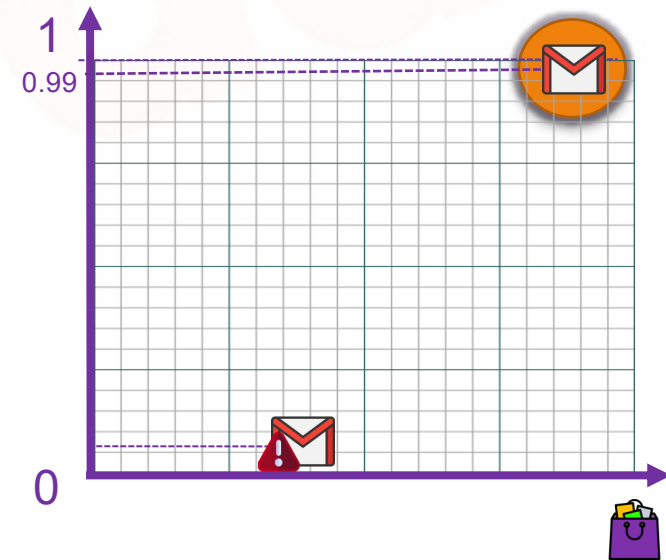


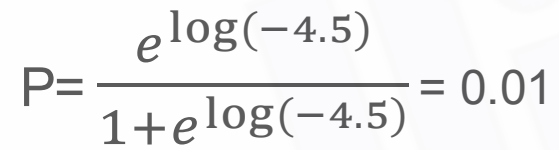


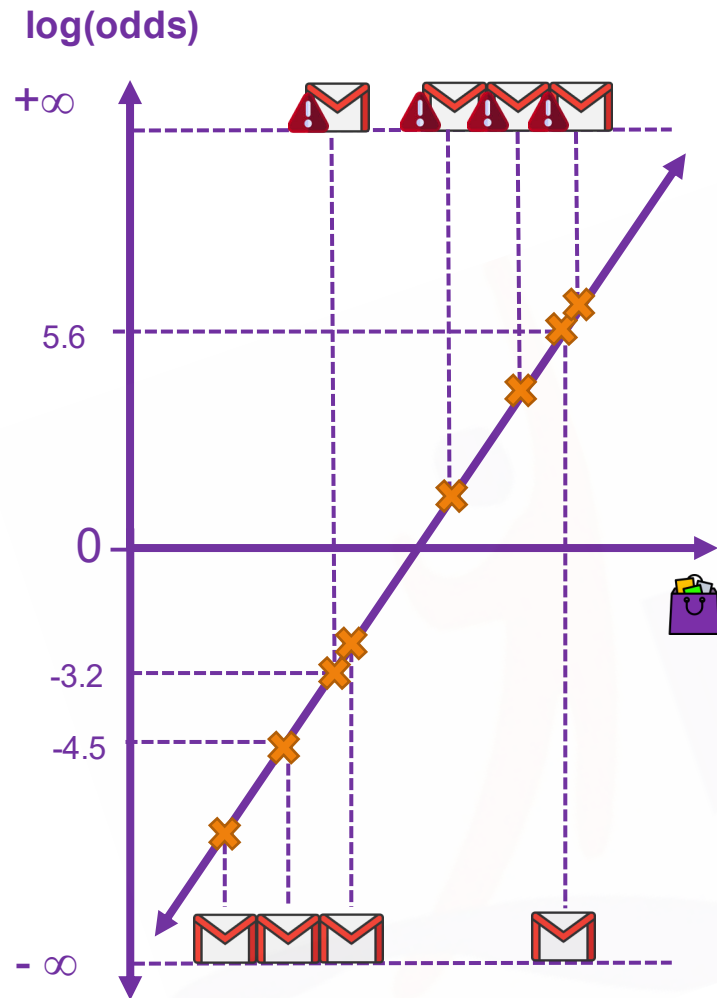
$$P = \frac{e^{\log(5.6)}}{1 + e^{\log(5.6)}} = 0.99$$



Probability of mail being spam



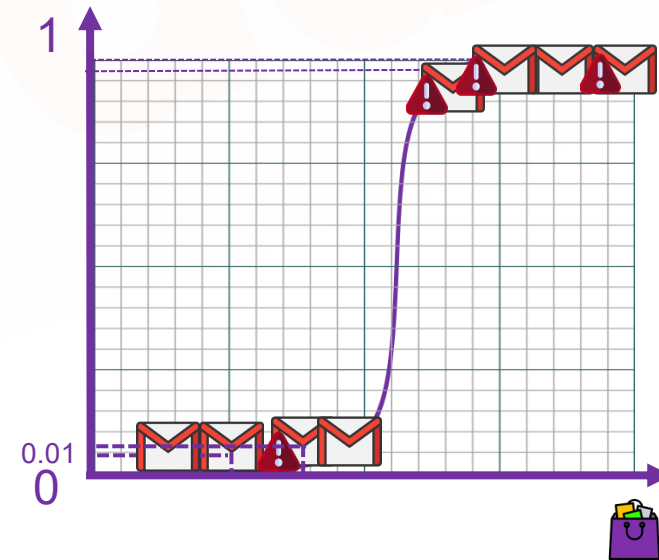




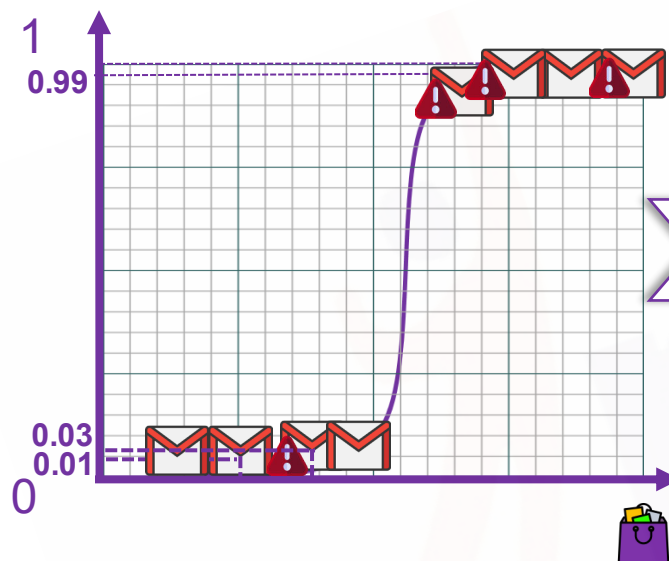
$$P = \frac{e^{\log(\text{odds})}}{1 + e^{\log(\text{odds})}}$$



Probability of mail being spam



Probability of mail being spam



Now let us find out the individual log likelihood of each mail

Likelihood of 1st mail being spam = 0.01

Likelihood of 2nd mail being spam = 0.01

Likelihood of 3rd mail being spam = 0.03

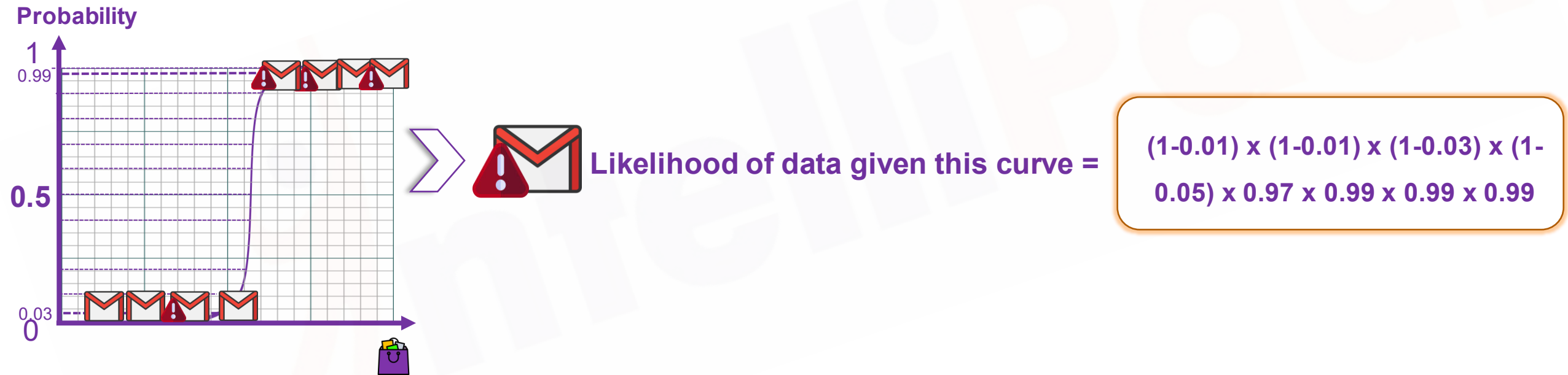
Likelihood of 4th mail being spam = 0.05

Likelihood of 5th mail being spam = 0.97

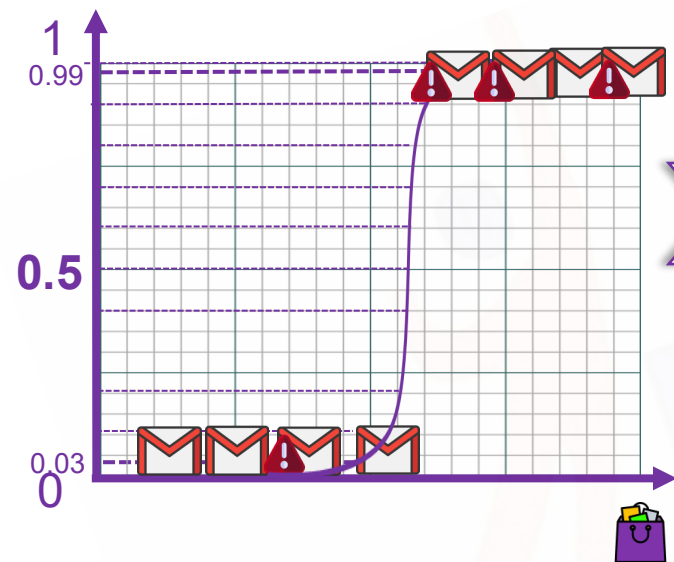
Likelihood of 6th mail being spam = 0.99

Likelihood of 7th mail being spam = 0.99

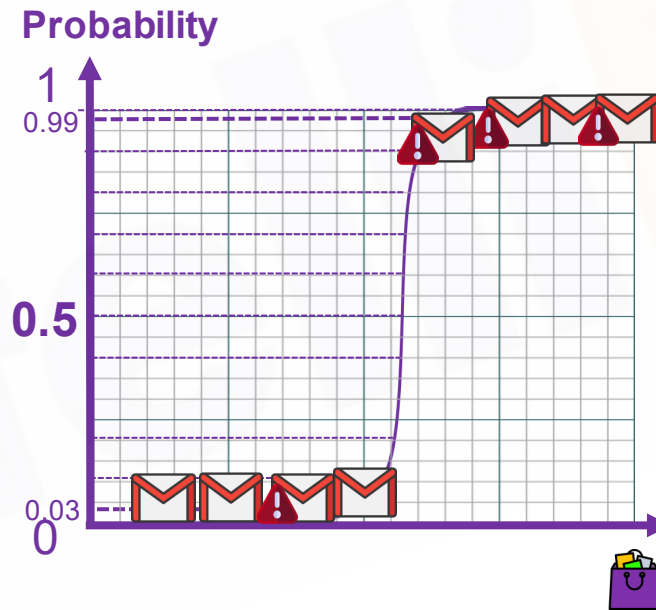
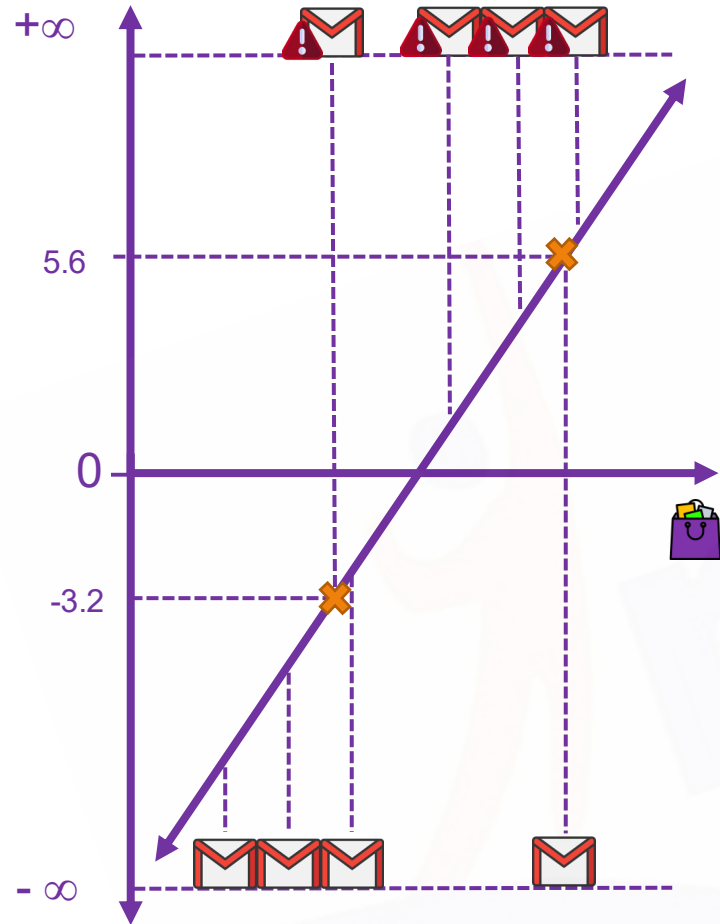
Likelihood of 8th mail being spam = 0.99



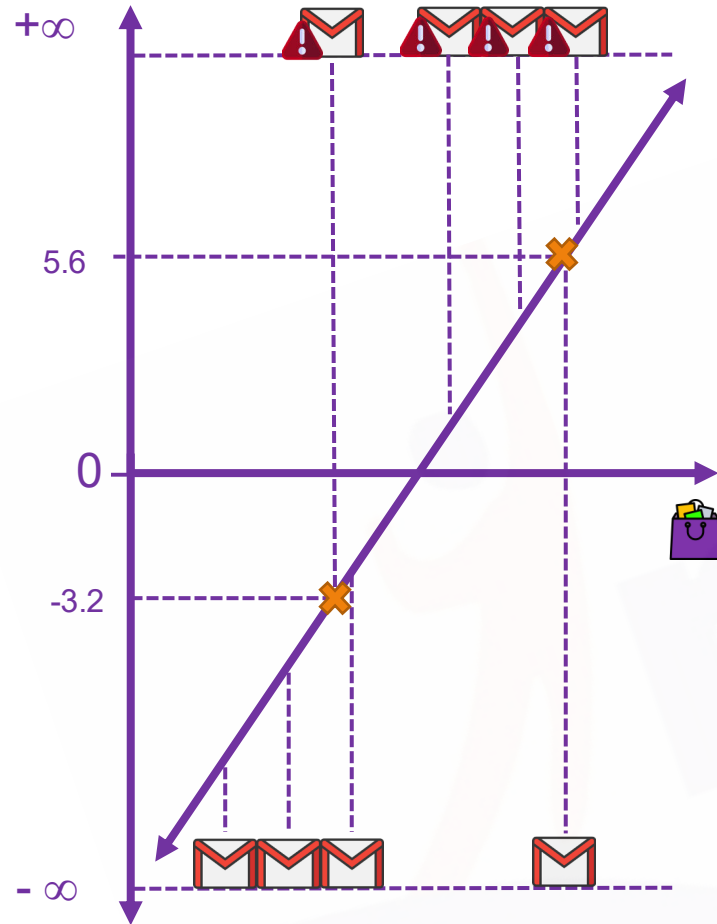
Probability

 $\log(\text{Likelihood of data given this curve}) =$

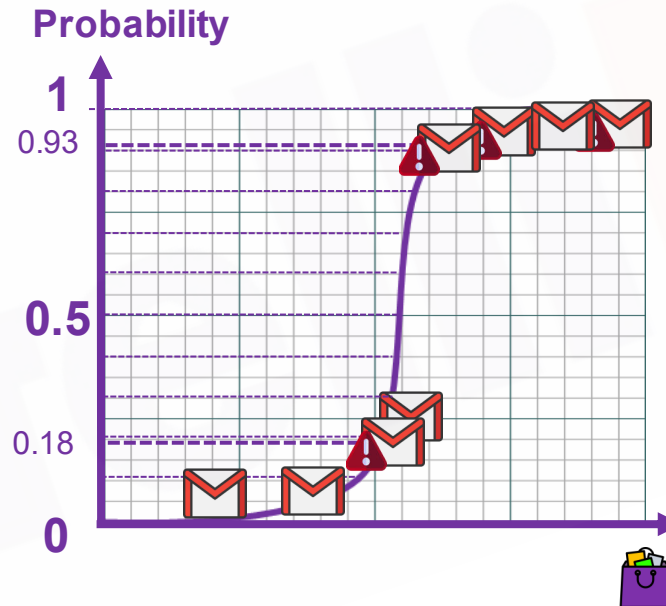
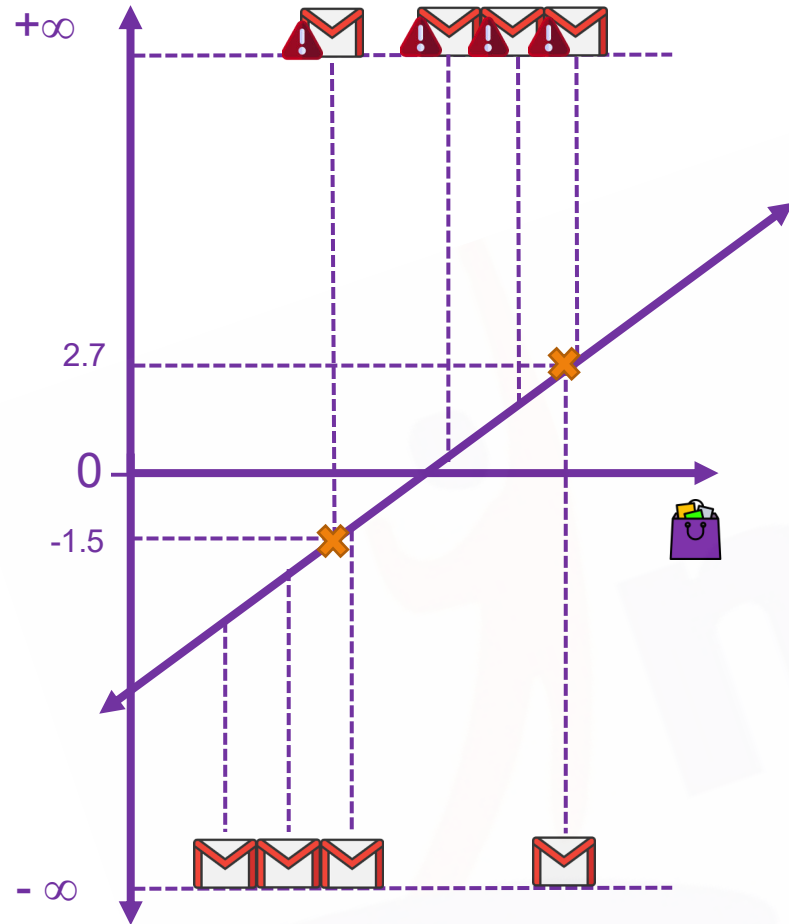
$$\begin{aligned} & \log(1-0.01) + \log(1-0.02) \\ & + \log(1-0.05) + \log(1-0.05) \\ & \quad -0.084 \\ & + \log(0.97) + \log(0.99) \\ & + \log(0.99) + \log(0.99) \end{aligned}$$



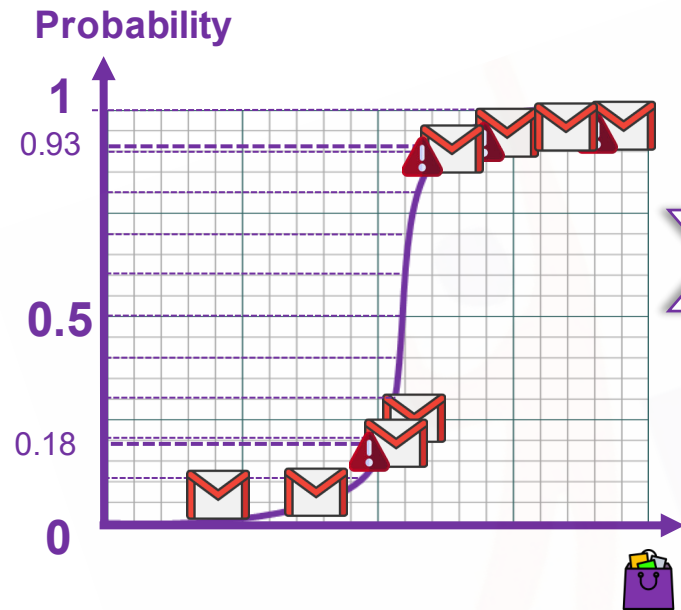
This means that log likelihood of this line is **-0.084**



Now let us rotate this line to find out the best fitted regression line.

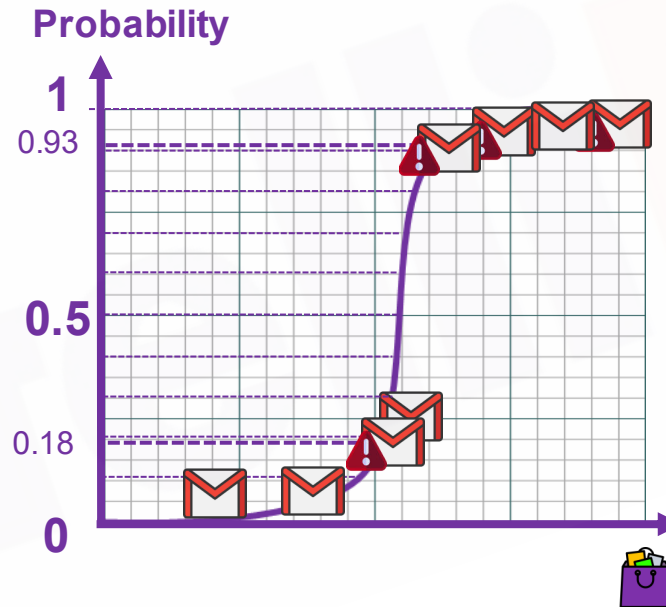
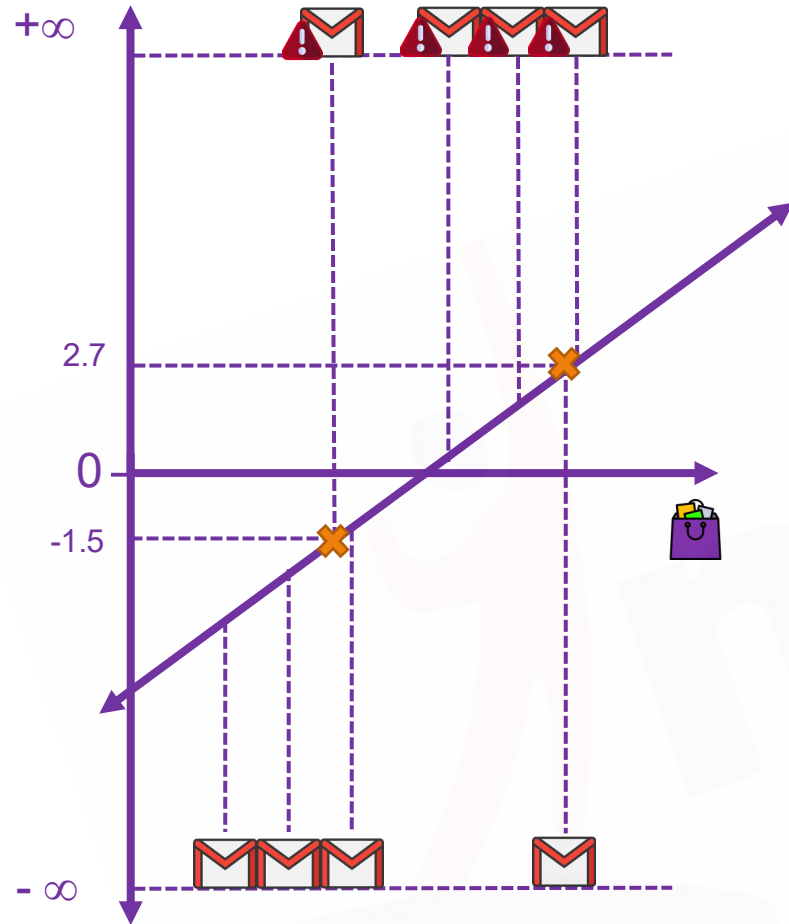


Again we will calculate individual log likelihood of each mail

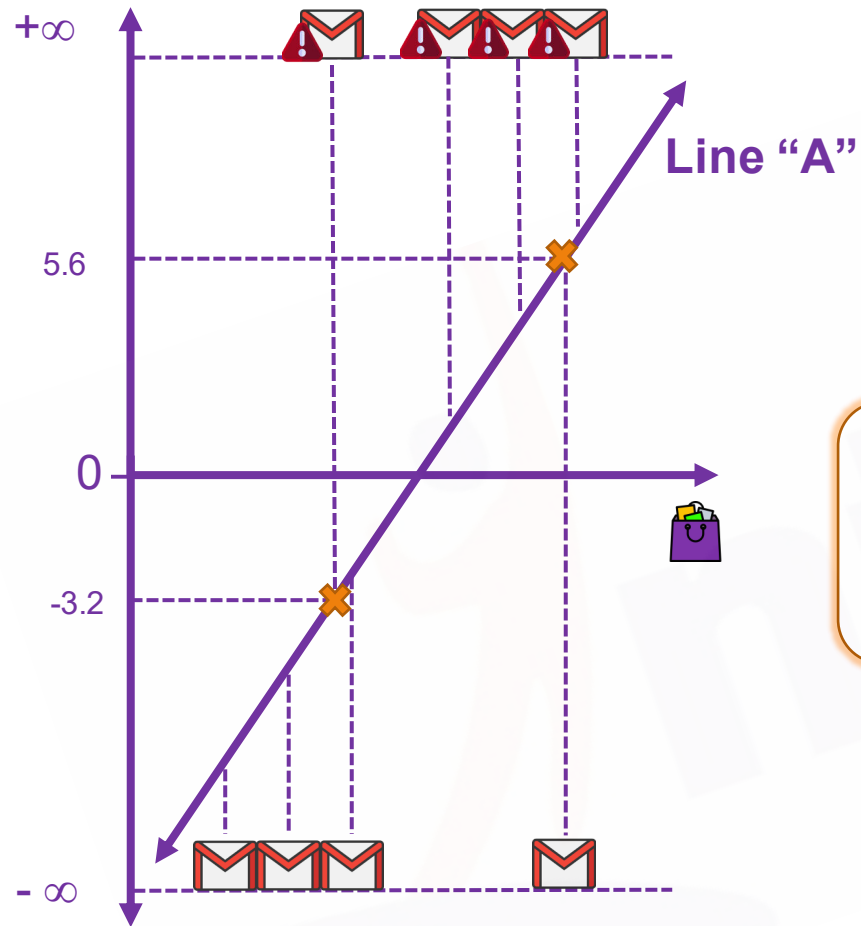


log(Likelihood of data given this curve) =

$$\begin{aligned} & \log(0.93) + \log(0.97) \\ & + \log(0.99) + \log(0.99) \\ & + \log(1-0.18) + \log(1-0.2) + \\ & \log(1-0.03) + \log(1-0.03) \end{aligned}$$

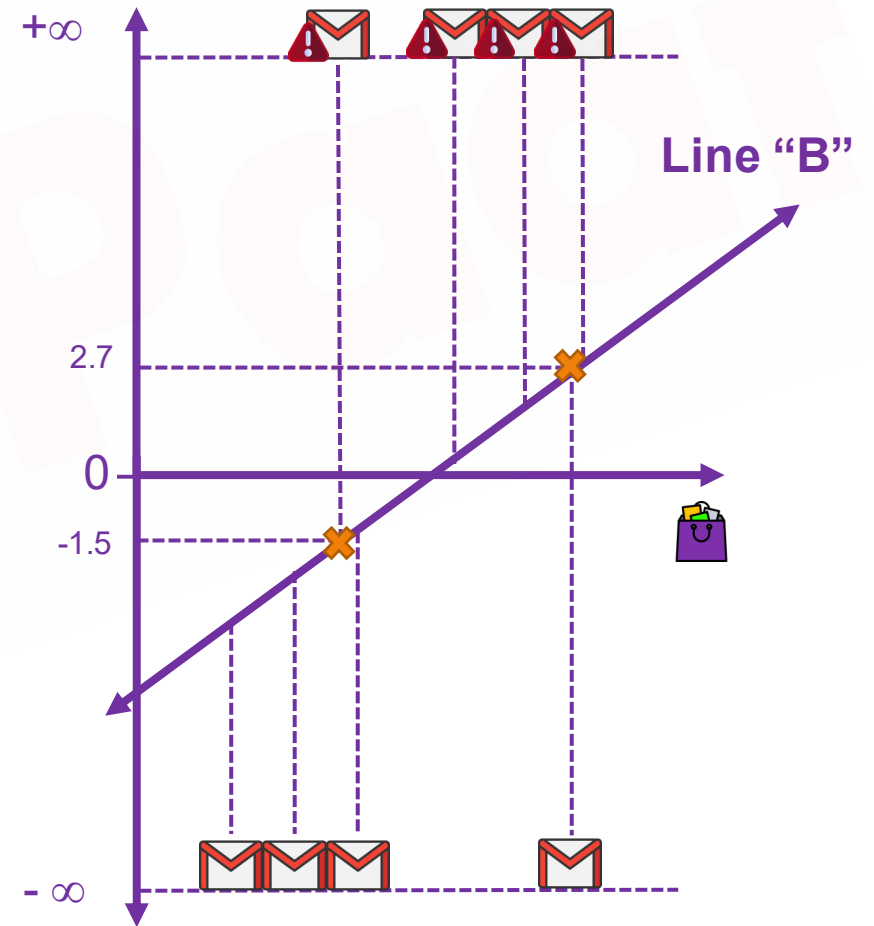


This means that log likelihood of this line is -0.207

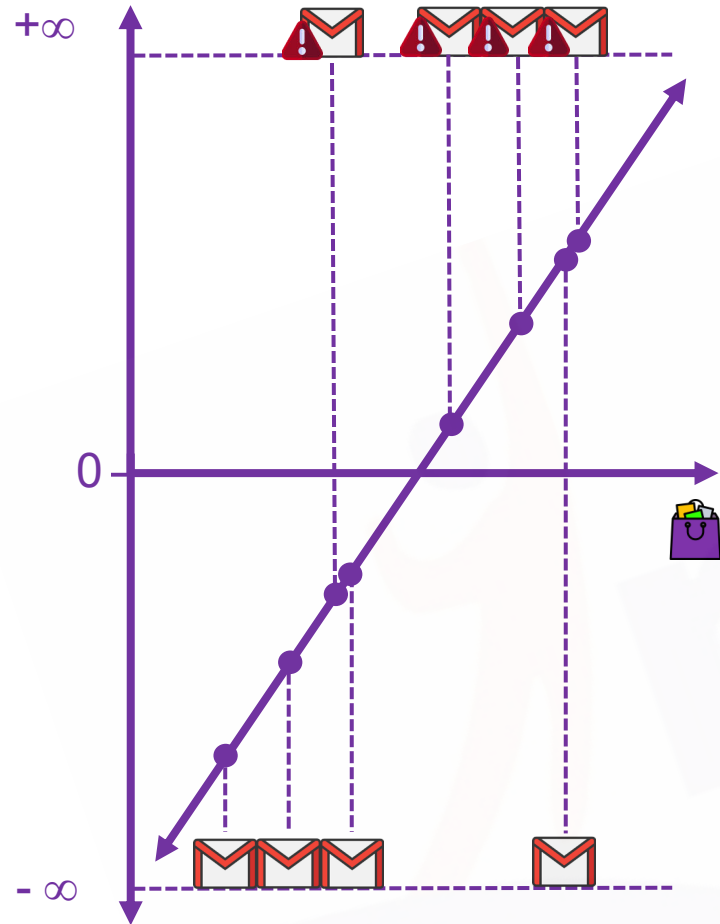


$$\log(\text{likelihood}) = -0.084$$

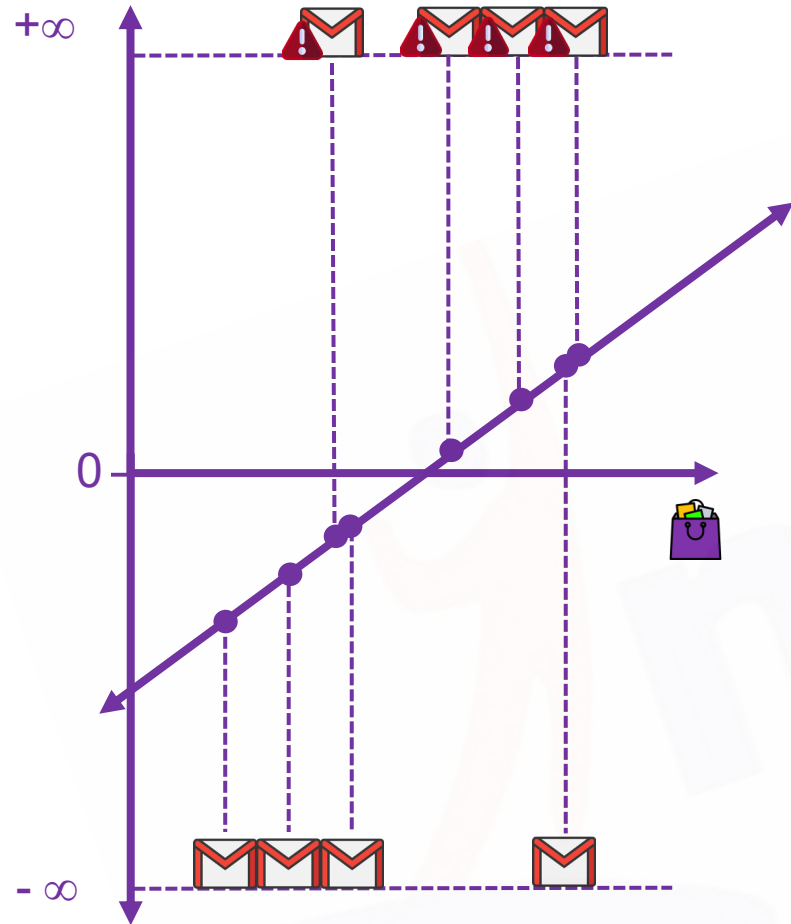
Line "A" has a better likelihood value than line "B"



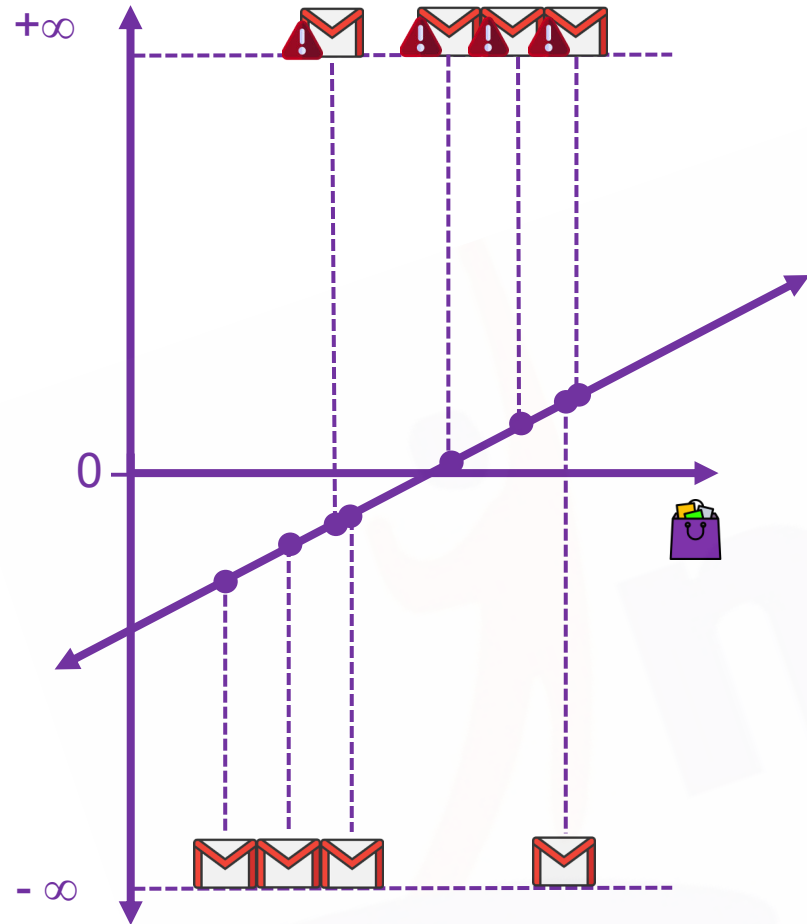
$$\log(\text{likelihood}) = -0.207$$



Again we will rotate the line

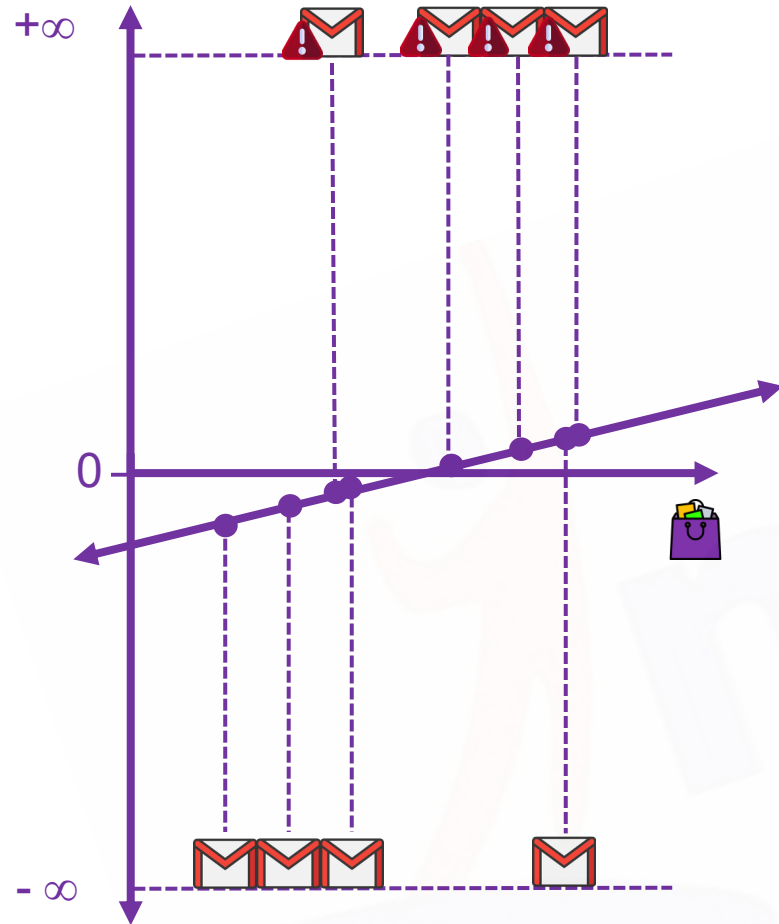


We will keep on rotating the line



We will keep on rotating the line

Until we get the maximum log likelihood



We will keep on rotating the line until we get the maximum log likelihood

That would be the best fitted regression line

**Look back of Linear
Regression**

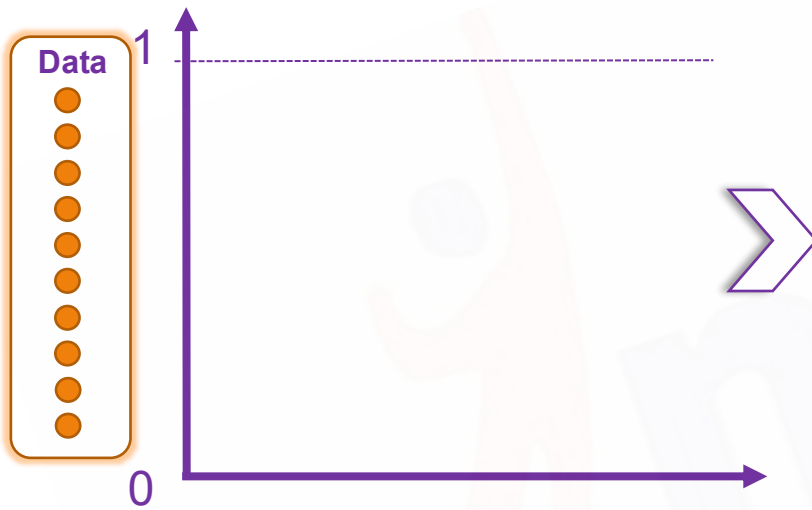
**Introduction to Logistic
Regression**

**What and Why logistic
regression?**

Summary

A

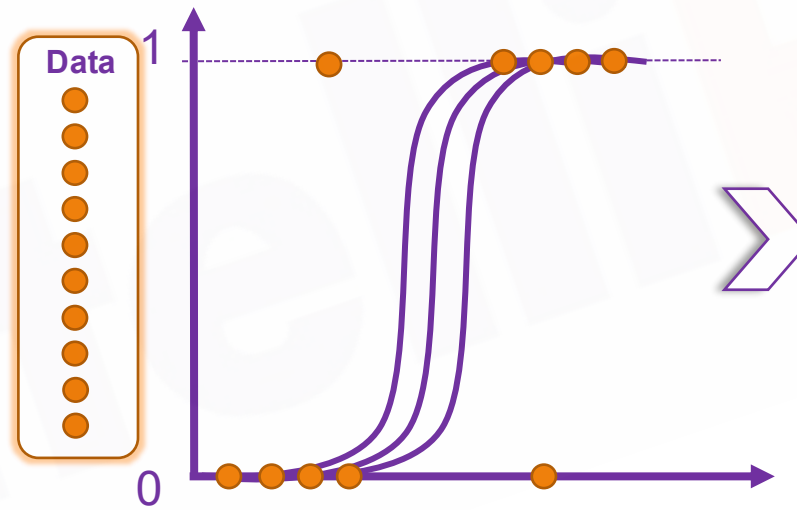
Plot the labeled data



Log(odds)

B

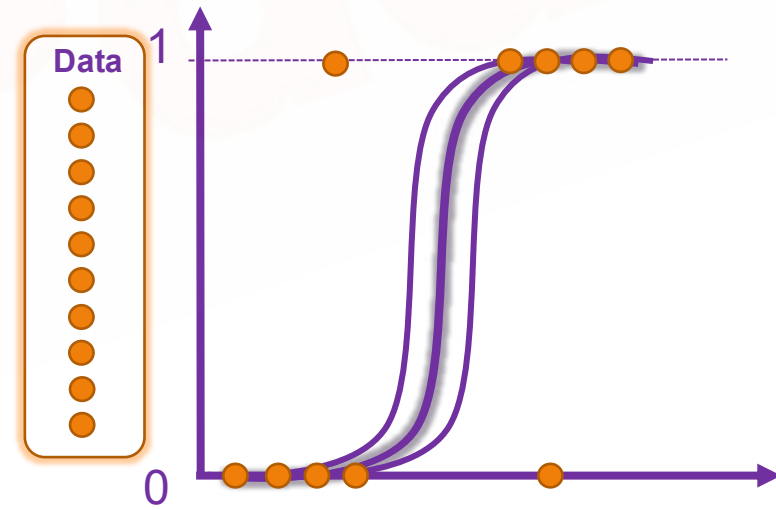
Draw a regression line and keep rotating



Sigmoid
Function

C

Use MLE to find out the best fitted regression line



MLE

Thank
You



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